

A comparison between copula-based, mixed model, and estimating equation methods for regression of bivariate correlated data

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Abstract: This paper presents a simulation study comparing the performance of generalized joint regression models (GJRM) with generalized linear mixed models (GLMM) and generalized estimating equations (GEE) for regression of longitudinal data with two measurements per observational unit. We compare models on the basis of overall fit, coefficient accuracy, and computational complexity.

We find that for the normal model with identity link, all models provide accurate estimates of regression coefficients with comparable fit. However, for non-normal marginal distributions and when a non-identity link function is used, we highlight

a major pitfall in the use of GLMMs: without significant adjustment they provide highly biased estimates of marginal coefficients due, inter alia, to estimand mismatch, and often provide unsatisfactory distributional fits. The bias of the GLMM estimate and its relative lack of fit is more pronounced when the marginal distributions are more skewed or highly correlated. In addition, we find major discrepancies between the estimates from different GLMM software implementations. In contrast, we find that GJRM and GEE provide accurate marginal coefficient estimates and standard errors across the simulated settings, with GJRM overall fit being generally better than GLMMs by likelihood-based criteria and comparable to GEE by variogram score. We also extend the results to trivariate Bernoulli data and compare the approaches for a real-world longitudinal study of doctor visits.

We conclude that for non-normal bivariate data, the GJRM provides a strong likelihood-based alternative with more consistently interpretable marginal coefficients than the GLMM, and generally better overall fit. The GJRM's performance is typically comparable to the GEE, while providing additional flexibility in choice of marginal distribution, likelihood-based tools such as AIC/BIC and control over correlation structure.

Key words: copula; correlated data; mixed models; generalized joint regression; random effects

1 Introduction

Dependence between random variables often develops when repeated observations are taken from a sampling unit, for example, cholesterol levels over time for a single patient. If independence is assumed between these correlated observations, the obser-

vations will exhibit a lower than expected variance, which causes bias and inefficiency in coefficient estimation (Laird and Ware, 1982).

Several approaches have been developed to date to account for the dependence between observations in regression, the most popular in the case of non-normal data being generalized linear mixed models (GLMM) (Breslow and Clayton, 1993) and generalized estimating equations (GEE) (Liang and Zeger, 1986). These methods introduce an adjustment to univariate methods that account for the dependence structure between observations. Both frameworks have been substantially extended since their inception to apply to a much broader range of response distributions, as well as to incorporate smooth terms for covariates (Hastie and Tibshirani, 1990; Stasinopoulos et al., 2024; Yee and Wild, 1996; Wild and Yee, 1996).

Both the GEE and GLMM are based on the framework of the Generalized Linear Model (GLM) (McCullagh and Nelder, 1989) and introduce dependence to the GLM via different methods. For a given longitudinal dataset y_{it} observed at times $t = 1, \dots, T$ and for subjects $i = 1, \dots, n$, the standard GLM assumes within-subject (and between-subject) independence. For the GLM, we have the model:

$$g(\mu_{it}) = \eta_{it} = x_{it}^T \beta$$

where $g(\cdot)$ is the link function that connects the linear predictor η_{it} to the response parameter μ_{it} and β is the parameter vector that links the covariates x_{it}^T to the linear predictor.

The GEE uses the same model base as the GLM but introduces a working correlation matrix among measurements for a subject (or cluster) i , R_i , with assumed independence between subjects. This correlation structure is used to provide adjusted

population and individual-level standard errors of parameter estimates to account for correlation. Given the flexibility of the covariance structure, multiple structures can be modeled, popularly: a single correlation parameter ρ for all observations in a cluster, referred to as an exchangeable correlation structure; AR1 with a single correlation parameter but with correlation decaying with increasing distance; or fully unstructured. Modern GEE software implementations ([Vanegas et al., 2023](#)) extend the available correlation structures further and provide additional tools for model selection, diagnostics, and more complex datasets or covariates. This method provides accurate estimates for marginal coefficients while providing more accurate standard errors than the comparable GLM for a correlated dataset. Importantly, one significant drawback of the GEE is that it does not utilize a valid likelihood, so resulting estimates are not maximum likelihood, and standard likelihood-based model selection criteria such as Akaike Information Criterion (AIC) are not available.

The GLMM introduces dependence between subjects by introducing random intercept and/or random slope terms into the linear predictor, which provide a common value for all observations of a given subject. The simplest and most commonly used form of the GLMM is that of a single random intercept term that induces dependence within subjects by introducing a normally distributed random effect term centered at zero, $b_i \sim \mathcal{N}(0, \theta^2)$, $i = 1, \dots, n$, into the linear predictor of the GLM, which is fixed across all observations for subject i :

$$g(\mu_{it}) = \eta_{it} = x_{it}^T \beta + b_i.$$

The introduction of the random effect term changes the structure of the data being modeled as instead of the distributional model directly modeling the marginal response distributions as in the case of the GLM and GEE, i.e. $Y_t \sim \mathcal{D}(\mu_t, \sigma_t)$, the

distribution being modeled is conditional on the random effect, $Y_t|b \sim \mathcal{D}(\mu_t, \sigma_t)$.

In the case of a mixed model with a random effect term, the way parameters are interpreted needs to be carefully considered due to the impact of the random effect on the parameter estimates of the model (Gory et al., 2021). This is because there is a difference between the marginal and conditional estimates in the GLMM framework. For example, if the means of the marginal distributions are the values of interest, i.e. $E(Y_t)$, then when $g(\cdot)$ is the identity link function, the conditional and unconditional expectations for the means are equivalent because the expected value of the random effect is zero and can be dropped, i.e.

$$E_b[E_Y(Y_t|b_i)] = E_b [x_{it}^T\beta + b_i] = x_{it}^T\beta + E_b(b_i) = E(Y_t).$$

However, in the case where the link function is not identity, then $E_b[E_Y(Y_t|b_i)] \neq E_Y(Y_t)$ as the link function prevents the random effect from being dropped and needs to be evaluated over the random effect, i.e.

$$E_b[E_Y(Y_t|b_i)] = E_b [g^{-1}(x_{it}^T\beta + b_i)] \neq E(Y_t),$$

in general.

This means both the shape of the distribution being modeled and the parameter estimates from the GLMM are distorted by the random effect, and as the parameters are based on the conditional expectation this makes interpretation problematic. This issue of parameter interpretation exists across all popular non-identity link functions (Gory et al., 2021).

Many solutions have been proposed to the problem of providing better marginal interpretability for GLMMs. These solutions include: specifying the models for marginal

interpretation, termed marginalized multi-level models, by incorporating an additional adjustment term in the linear predictor for each observation (Gory et al., 2021; Heagerty and Zeger, 2000; Mills et al., 2002); the use of bridging distributions such that the integral of the predictor $h(\eta + b)$ with respect to the bridging distribution provides marginal parameters (Wang and Louis, 2003); and selecting a random effect distribution which is conjugate to the outcome distribution which can provide closed form solutions to estimates for marginal joint distributions and marginal parameters for many exponential family distributions (Molenberghs et al., 2010). Helpful overviews of the relationship between marginal, bridging, and conjugate solutions are provided by Kenward and Molenberghs (2016) and Molenberghs et al. (2013). These methods generally introduce additional restrictions to analysis, for example, restricting to only exponential family distributional forms, and complexity to the construction and estimation of the model, largely due to the need for integration over often complex random effect components and the adjustments to the model structure to account for them.

In practice, these adjustment methods are rarely applied; rather, the simplifying assumption $E(Y_t|b_i) = E(Y_t)$ is used, generally without clarification, and inference is based on the results of standard packages that provide conditional results. For this reason, in this paper we analyze the results of a standard GLMM implemented by common packages without the major adjustments set out above. To highlight this point of how commonly GLMMs are used without adjustment, we performed a non-comprehensive review of recent papers utilizing GLMMs in the life sciences. We used the search term "generalized linear mixed models" in PubMed and reviewed the 20 most recent papers of 252 published in 2024, resulting in papers with creation dates from October 26 to November 21, 2024. Our interest is in the total volume

and breadth of the articles, which is significant, the complexity of the structure used, and whether complex adjustments are made to account for the difference between marginal and conditional results.

Of the twenty papers reviewed, eight were non-normal GLMMs with a single random intercept. Of the others, two had two or more random effects, two were linear models (normal response with identity link), two were meta-analyses, three were not studies including GLMMs, and three had more complex structures. None applied the complex adjustments outlined above. The papers model a broad range of measures in health and life sciences including: association between 24-hour movement guidelines and academic engagement in adolescents ([Sánchez-Miguel et al., 2024](#)), sex differences in survival following acute coronary syndrome ([Anand et al., 2024](#)), fetal risk for congenital heart defects ([Pi et al., 2024](#)), increasing HIV testing uptake using mobile technology ([Schnall et al., 2024](#)), novel soft contact lenses impact on patient comfort ([Buch et al., 2024](#)), antenatal care utilization in Zambia ([Shumba et al., 2024](#)), negative affect and loss of control eating following bariatric surgery ([Kerver et al., 2025](#)), and the association between residential greenness and obesity ([Yu et al., 2024](#)). More broadly, there are applications across life sciences, including nutrition ([Müllertz et al., 2024](#)), medicine ([Rysavy et al., 2020](#); [Chopra et al., 2017](#)) and fisheries ([Gundelund et al., 2022](#)).

Many authors highlight the rapidly increasing use of GLMMs to analyze correlated data in life sciences research papers over the past 20 years and provide reviews of their use specifically in clinical medicine ([Casals et al., 2014](#)), psychology ([Bono et al., 2021](#)) and biology ([Bolker et al., 2009](#); [Pekár and Brabec, 2018](#); [Thiele and Markussen, 2012](#)). Interestingly, while the GEE provides a similar adjustment for

correlation, it has not seen anywhere near the increase in adoption as GLMMs (Pekár and Brabec, 2018). In the vast majority of cases, the GLMM is applied simply for the purpose of adjusting for known correlation between observations within clusters, that is, to adjust fixed effect parameters, essentially treating the correlation as a nuisance parameter (Thiele and Markussen, 2012; Pekár and Brabec, 2018). Most of the studies in these reviews use a single random intercept (Casals et al., 2014; Pekár and Brabec, 2018), which is why we focus on this structure throughout this paper. The most popular model error structures used in the reviewed papers were Poisson, Binomial, and Bernoulli.

The increased complexity of the GLMM structure introduces additional risks to the modeling process that are difficult to account for even for an experienced analyst (Silk et al., 2020; Bolker et al., 2009; Arnqvist, 2020). For example, one analysis (Bolker et al., 2009) finds 58 percent of reviewed papers used GLMM inappropriately in some way, and others (Casals et al., 2014; Bono et al., 2021) note that the majority of the articles reviewed did not report critical model information needed to assess the validity of the model, or even the distribution and link function in many cases.

1.0.1 Copula regression

More recently, approaches for joint modeling of multivariate dependent data in a regression framework have emerged through the use of copulas, which provide an alternative to the former models. In particular, generalized joint regression models (GJRM) (Marra and Radice, 2017) provide a likelihood-based framework to jointly model two or three (with more limited distributions) correlated variables and their dependence structures while incorporating covariate effects. The GJRM combines the utility of the likelihood-based approach of the GLMM with the marginal inter-

pretability of the GEE.

The GJRM framework relies on the use of copulas, which provide a convenient method for deconstructing a bivariate distribution into a combination of two marginal distributions and a copula function to model the dependence structure (Nelsen, 2007; Trivedi and Zimmer, 2007). Any continuous bivariate distribution can be represented by a unique combination of two marginal distribution functions and a copula function, and this extends readily to the multivariate case (Sklar, 1973). Multivariate discrete distributions can also be deconstructed in the same way with copulas; however, the resulting distribution is not unique, and additional considerations are required to ensure uniformity of transformed margins when fitting copulas (Rizal et al., 2021). Copulas have been widely used in economics and financial time series analysis, particularly in portfolio risk (Palaro and Hotta, 2006; Pitt et al., 2006) and insurance loss estimation (Krämer et al., 2013).

One of the key developments in copula regression is the introduction of a framework for conditioning a copula on a variable (covariate) (Patton, 2006). This allows dependence structures to be modeled after the effects of covariates are removed, leaving the true residual dependence structure and significantly expanding the applications of copulas in a regression context. This method is referred to as conditional copula regression. A large amount of literature has focused on the development of likelihood-based tests for the existence and significance of covariates in a conditional copula regression framework (Acar et al., 2011; Gijbels et al., 2011; Acar et al., 2013; Craiu and Sabeti, 2012; Sabeti et al., 2014). This results in a highly flexible overall modeling framework for correlated data whereby the marginal distributions and the copula function can, in theory, both have fitted shapes and both rely on any number or type

of covariates.

For modeling correlated data with two clusters or time points, there is a broad range of defined copula functions available to fit many different bivariate dependence structures. For example, the Clayton copula can capture varying levels of lower tail dependence.

To model a dependence structure with more than two time points or clusters requires the definition of a multivariate copula; available options are highly limited. However, a popular solution to the problem of defining a multivariate copula is the use of pair-copula constructions. These use tree-based representations of multiple bivariate copulas to deconstruct a T -dimensional multivariate copula into $T(T - 1)$ bivariate copulas (Czado and Nagler, 2022). This allows for the use of the broad range of available bivariate copula shapes to model multivariate dependence.

1.0.2 *Software*

The first implementations for fully flexible joint regression with regression covariates for both the margins and dependence are limited to modeling the bivariate case and, for some specific distributions, trivariate. These approaches are the Generalized Joint Regression Model (GJRM) (Marra and Radice, 2025) and the Generalized Additive Model for conditional dependence structures (gamCopula) (Vatter and Chavez-Demoulin, 2015). Both approaches combine univariate models for marginal distributions, generalized additive models for location, scale, and shape, GAMLSS (Stasinopoulos et al., 2024), and generalized additive models, GAM (Hastie and Tibshirani, 1990), respectively, with a fit for a copula to capture the dependence structure. The methods differ in their approach to optimization of the joint likeli-

hood function: the **R** (R Core Team, 2024) package **GJRM** (Marra and Radice, 2017) uses a simultaneous estimation approach for the joint likelihood of the copula and marginal distributions, and **gamCopula** (Vatter and Chavez-Demoulin, 2015) is used in a two-stage approach where marginal and copula likelihoods are optimized separately. The **GJRM** package supports many, but not all, of the broad range of marginal distributions provided by **gamlss** (Rigby and Stasinopoulos, 2005). In our analysis, we use **GJRM** in preference to **gamCopula** as the implementations provide similar results for the copula parameters, but **GJRM** delivers slightly less biased and more efficient estimates for marginal parameters than **gamCopula**, and has faster computation (Marra and Radice, 2017). The joint likelihood approach of GJRM also directly calculates changes in marginal coefficient estimates and SEs as part of optimization, which is critical for comparison to existing methods which jointly estimate all coefficients. In addition, the GJRM implementation has been substantially extended since its development, with detailed support published (Marra and Radice, 2025). A Bayesian approach for simultaneous estimation in a bivariate copula regression has also been introduced; this provides another alternative to the former methods (Klein and Kneib, 2016; Kock and Klein, 2025), however, we focus this analysis on likelihood- and quasi-likelihood methods due to their relative popularity and computational advantages. Future reviews may focus on further comparisons.

Another popular package used in copula-based modeling is **VineCopula** (Czado and Nagler, 2022), which provides a method for fitting multivariate dependence structures by combining bivariate copulas, termed pair-copula constructions. This dependence model is fit to the residual from a fitted model for each margin, for example, a GLM. This method is referred to as a two-stage method for multivariate regression, as opposed to a one-stage joint model which fits margin and dependence simultane-

ously. This method has been extended to allow each of the copula parameters in a pair-copula construction to depend on any number of covariates using the package **gamCopula** and is termed Generalized Additive Models for Pair-Copula Constructions (Vatter and Nagler, 2018). However, since the two-stage approach does not jointly fit copula and margin, as is the case for the GJRM, it does not provide adjustments to marginal parameter coefficients and SEs as part of the fit. This is a substantial disadvantage of the method in comparison to the joint approach of the GJRM, as the marginal coefficients and standard errors are generally the parameters of most interest in longitudinal regression, as opposed to the correlation parameters.

It is possible to estimate improved marginal parameter standard errors using MCMC simulation of the marginal and VineCopula two-stage model. However, this method is computationally slow and will not allow for true adjustment of marginal coefficient estimates to the best overall fit due to the lack of joint likelihood and the margin being fixed as part of the estimation method. There are also joint Bayesian approaches to estimating marginal parameter uncertainty using credibility intervals (Czado et al., 2010), however these are not directly comparable to existing methods, and as noted previously, we focus on likelihood-based approaches in this paper. This is the reason why the joint likelihood approach of the GJRM approach is superior for longitudinal regression and, we argue, requires extension to greater dimensions, as it provides an optimal fit for the true joint distribution and directly estimates marginal parameter coefficients and SEs which are the most critical components of studies reviewed.

1.1 Contribution and approach

Adjustments for dependence incorporated in GLMMs and GEEs are essentially approximate approaches for accounting for multivariate data features in a univariate regression framework. In contrast, a copula-based joint regression approach directly models the multivariate distribution, with fits for observed margins and dependence structure and all relevant coefficients. This enables a high level of control and transparency in describing the underlying data. The structure also allows for a high level of flexibility in use of the final model, for example, being able to analyze the fitted tails of the margin or copula distribution.

To date, applications of copula-based joint regression models that use covariate-dependent copulas and margins have been mainly limited to the bivariate and trivariate case and have focused mainly on modeling different outcome variables simultaneously, for example, simultaneous modeling of poverty and leisure time ([Marra et al., 2023](#)) or health expenditure and health outcomes ([Dorn et al., 2023](#)). We review an alternative potential use for copula-based joint regression models for the case of longitudinal data regression, where the same outcome variable is measured at multiple time points for the same cluster or subject. We are not aware of any previous research that has used copula-based joint regression models for longitudinal data regression and compared them to existing methods.

This paper focuses on contrasting the performance of copula-based joint regression models, in particular the generalized joint regression model (GJRM) ([Marra and Radice, 2025](#)), with the most popular methods for adjusting for dependence in regression: the incorporation of a single random intercept term in standard generalized linear mixed models (GLMM), and generalized estimating equations (GEE).

We limit our investigation to the simplest case of the estimation of coefficients for two correlated outcome variables having the same distributional form dependent on covariates. This corresponds to a longitudinal model with two time points, so there are two measurements per observational unit.

The restriction to bivariate and Bernoulli trivariate data is due to the following three reasons. Firstly, and primarily, due to the limitation of the **GJRM** software, which provides the only methods for bivariate and, for a limited set of distributions, trivariate joint regression. While methods exist for multivariate fits using two-stage approaches, for example, fitting a marginal model then a VineCopula copula-structure to the residuals, these methods are not directly optimized for analysis of marginal coefficients as the parameters of interest, as these are calculated by the marginal model already included as comparisons in our analysis. Secondly, in two dimensions we have substantially greater control over the correlation structure, removing the complexity of multiple layers of dependence which allows for a fairer comparison to existing approaches which do not provide the same level of control over dependence structures as multivariate-copula based regression. For a distribution with 10 time points for example, a pair-copula construction would create a fit for $T(T-1)/2 = 45$ copulas, which will naturally perform well compared to simpler correlation adjusting models if correlation is highly heterogeneous between time points, but will be severely disadvantaged if the correlations are more similar between pairs. Third, and perhaps most importantly, as noted previously, it is possible to estimate adjusted marginal parameter coefficients and standard errors for VineCopula based two-stage models to some extent using MCMC simulation, however, this method is manual and highly computationally inefficient compared to extracting these results directly from the joint fit. The manual nature of the approach, and its inefficiency means it is not

a reasonable alternative to simple existing methods for a practitioner, particularly given the much simpler GLMM approaches are already often used incorrectly.

1.2 Overview of analysis

We structure our analysis in three parts:

1. **Broad / exploratory investigation for simulated bivariate data fits:**

First, we analyze the simplest case of simulated bivariate data with covariates with a broad range of marginal distributions, dependence structures and parameter combinations to identify key areas of contrasting performance between models.

2. **Confirmatory investigation for trivariate data fits:** We then extend our simulations to trivariate Bernoulli data focusing on confirming whether findings do or do not extend from the bivariate case.

3. **Applications data:** All bivariate models are applied to a real-world dataset of annual doctor visits over time, a zero-inflated count distribution.

For the exploratory analysis of bivariate distribution fits, where possible, we simulate bivariate distributions which originate from neither random effects nor a copula, so as not to bias the estimation towards either of these models. We provide a detailed overview of these distributions in Appendix A. The complete set of simulated bivariate distributions and parameter ranges is included in Table 1.

For both bivariate and trivariate fits, we compare model performance of copula-based regression models with the GLM, GEE and different GLMM software on the basis of:

Distribution	Gaussian	Gamma	Negative Binomial	Bernoulli
Type	Continuous	Continuous	Discrete	Discrete
Value range	Unbounded	Positive	Positive	Binary
Dependence	Normal	Low-value skew	Low-value skew	Normal
Link Function	Identity	Log	Log	Logit
Time points	2	2	2	2
Observations per time point	1,000	1,000	1,000	1,000
Simulated Parameter Range				
μ_1	1	(2, 21)	(5^{-3} , 5^3)	(0.1, 0.9)
μ_2	2	$1.2\mu_1$	(5^{-3} , 5^3)	(0.1, 0.9)
σ_1	(0.25, 2.5)	($\sqrt{0.2}$, $\sqrt{2.1}$)	(5^{-1} , 5)	—
σ_2	(0.25, 2.5)	($\sqrt{0.2}$, $\sqrt{2.1}$)	(5^{-1} , 5)	—
ρ	(0.1, 0.9)	—	—	(0.1, 0.9)
Covariate Coefficients				
$x_1 \sim \text{Binary}$	1	1	1	1
$x_2 \sim \text{U}(1, 100)$	0.01	0.01	0.01	0.01
Total Parameter Combinations	225	400	400	125

Table 1: Bivariate simulation parameters: Overview of simulation parameters used for simulations of bivariate correlated data for comparison of model fits including distribution and link functions, simulated range of parameters and coefficients.

1. Accuracy of coefficient estimates and standard error,
2. Overall fit by likelihood-based model selection criteria and variogram scores (a proper scoring method), and
3. Computational complexity evaluated on runtime and convergence.

We demonstrate the following key points through this paper:

- For the bivariate normal distribution with identity link, all models (GJRM, GLMM, GEE, and GLM) provide accurate coefficient estimates and standard errors.
- For bivariate non-normal distributions, with non-identity link, and trivariate Bernoulli, the use of random intercepts in GLMMs to adjust for correlation, without major adjustment to model structure, provides highly biased estimates of the marginal intercept and covariate coefficients. This is due to the known estimand mismatch of the GLMM, which is often ignored in interpretation. This bias increases with greater marginal skewness and rank correlation. We also show substantially differing estimates of standard error and stability between standard GLMM implementations, including the **R** packages **lme4** (Bates et al., 2015), **gamlss** (Rigby and Stasinopoulos, 2005) and **mgcv** (Wood, 2017). In contrast, we show that the GJRM and GEE methods provide accurate and stable coefficient estimates and standard errors.
- We highlight that the GJRM provides overall model fit comparable to the GLMM by AIC and Generalized AIC (GAIC), and superior by Bayesian Information Criterion (BIC) to any other compared model. In addition, we show that the GJRM and GEE provide comparable and generally the best fits on the basis of variogram and weighted variogram scores compared to all other models when a non-identity link function is required.
- We demonstrate in a real-world dataset of counts of doctor visits over two years, which are zero-inflated and skewed, that the greater range of marginal distributions available to GJRM, which incorporates many, but not all, of the GAMLSS distributions, allows it to provide a substantially better model fit by likelihood criteria

than the non-GAMLSS GLMMs (**lme4** and **mgcv**); and a more easily interpretable model than the GAMLSS GLMM with the same marginal distribution. In addition, we show that variogram scores consistently select the GEE and GJRM over random-effect model alternatives for the dataset.

The key finding we highlight for the GLMM is a pitfall of simply applying a random intercept term to adjust for correlation without carefully accounting for the change in interpretation it implies. We suggest that the copula-based approach provides a more directly controllable and transparent approach to capturing correlation in bivariate correlated datasets. In general, the GEE and copula-based models provide comparable fits by other criteria, with the primary advantage of the GEE being substantially faster runtime, and the advantage of the GJRM being it provides an overall joint distribution, retains likelihood-based tools such as AIC, and the copula can depend on covariates.

2 Simulation

2.1 Model specification

For the purpose of model specification, we assume that the bivariate data are longitudinal with Y_1 and Y_2 being the random variables for the response at times $t = 1$ and $t = 2$, respectively. For the sake of generality of estimation, we assume marginals $Y_t \sim \mathcal{D}(\mu_t, \sigma_t)$ where $\mathcal{D}$ is a distribution appropriate to the class of model (exponential family for GLMs and GEEs; any distribution with computable derivatives for GAMLSS, and a large subset of GAMLSS distributions available in GJRM); where μ_t is generally the mean; and σ_t is a dispersion parameter. So, for the bivariate

distributions we model the following distributional parameters:

- μ_1 and μ_2 , the means of the marginal distributions,
- σ_1 and σ_2 , the dispersion of the marginal distributions, for the normal, Gamma, and negative binomial, and
- θ , the dependence parameter between the two marginals. This can be, for example, the Pearson correlation ρ , Kendall's τ , or the copula parameter.

In addition, we fit two covariates:

- x_1 , a binary factor with a value 0 or 1, with half of the individuals in each category, and
- x_2 , a uniformly distributed covariate between 1 and 100.

We estimate the parameters of four classes of models that approximate the underlying bivariate data:

- **GLM** is the simplest model included as a benchmark that assumes independence between time points:

$$Y_{it} \sim \mathcal{D}(\mu_t, \sigma)$$

$$g_\mu(\mu_{it}) = \beta_1 \mathbb{1}_{t=1} + \beta_2 \mathbb{1}_{t=2} + \beta_{x_1} x_1 + \beta_{x_2} x_2$$

$$\text{for } i = 1, \dots, n; \quad t = 1, 2.$$

The model is specified as above so that β_1 and β_2 are the exclusive parameters for times 1 and 2, respectively. Note that the GLM has a constant dispersion parameter, so we have $\sigma_1 = \sigma_2 = \sigma$.

- **GEE** incorporates an adjustment for dependence to the generalized linear model

through a correlation matrix. In this case with only two time points, there is only one correlation parameter, so there is no need to specify a correlation structure beyond this single parameter:

$$\begin{aligned}
 Y_{it} &\sim \mathcal{D}(\mu_t, \sigma) \\
 g_\mu(\mu_{it}) &= \beta_1 \mathbb{1}_{t=1} + \beta_2 \mathbb{1}_{t=2} + \beta_{x_1} x_1 + \beta_{x_2} x_2 \\
 R_i &= \theta \text{ is the correlation parameter} \\
 &\text{for } i = 1, \dots, n; \quad t = 1, 2.
 \end{aligned}$$

We use the simplest form of the GEE, which follows the distributional specification of the GLM, i.e., a constant dispersion parameter σ .

- **GLMM** in its simplest form with just a single random intercept combines a fixed linear model with a normally distributed random intercept term, which is assumed to be common to subjects between margins and induces dependence between observations at the two time points.

For the GLMM, the five parameters of the bivariate distributions can be captured directly. The most flexible model is the generalized linear mixed model with a random effect term and time-dependent σ :

$$\begin{aligned}
 Y_{it} &\sim \mathcal{D}(\mu_t, \sigma_t) \\
 g_\mu(\mu_{it}) &= \beta_1 \mathbb{1}_{t=1} + \beta_2 \mathbb{1}_{t=2} + \beta_{x_1} x_1 + \beta_{x_2} x_2 + b_i \\
 g_\sigma(\sigma_t) &= \beta_{\sigma_1} \mathbb{1}_{t=1} + \beta_{\sigma_2} \mathbb{1}_{t=2} \\
 b_i &\sim \mathcal{N}(0, \theta^2) \\
 &\text{for } i = 1, \dots, n; \quad t = 1, 2.
 \end{aligned}$$

We also consider the 4-parameter GLMM without a time-dependent dispersion parameter, i.e. where $g(\sigma_t) = \beta_\sigma$ for $t = 1, 2$; and an alternative model where

the random effect term is specified as having a nonparametric rather than normal distribution.

- **GJRM** jointly models the response variable at both times with a distributional fit and its dependence structure through a copula function to capture the full bivariate distribution:

$$\begin{aligned}
Y_{it} &\sim \mathcal{D}(\mu_t, \sigma_t) \\
g_\mu(\mu_t) &= \beta_1 \mathbb{1}_{t=1} + \beta_2 \mathbb{1}_{t=2} + \beta_{x_1} x_1 + \beta_{x_2} x_2 \\
g_\sigma(\sigma_t) &= \beta_{\sigma_1} \mathbb{1}_{t=1} + \beta_{\sigma_2} \mathbb{1}_{t=2} \\
\theta &\text{ is the copula parameter} \\
&\text{for } i = 1, \dots, n; \quad t = 1, 2.
\end{aligned}$$

In the GJRM framework (Marra and Radice, 2023), a distribution is selected for each of the margins and a copula function is selected to fit the dependence structure. For these simulated datasets, the marginal distributions are known. For the dependence structure, there are many varied copula shapes available for modeling. The Clayton, normal, Joe, Gumbel, and Frank copulas are included in the simulations.

For estimation we use the distribution which matches the marginals of the known bivariate distribution, e.g. the Gamma marginal distribution for the Bivariate Gamma. We select the identity link function for all models for the normal, the log link function for the negative binomial and Gamma, and the logistic link function for the Bernoulli.

For trivariate model fits, we specify an additional intercept parameter $\beta_3 \mathbb{1}_{t=3}$. We retain the intercept coefficients β_1 and β_2 and the two covariates x_1 and x_2 and their coefficients β_{x_1} and β_{x_2} in model specifications.

Model specification for the dependence structure for $T > 2$ requires additional consideration. For GLMMs, the same random intercept structure naturally extends to $T > 2$. For the GLM no adjustment is required, as the dependence between observations is ignored. For the GEE and copula-based method, the correlation structure specification extends from a single parameter to a parameter matrix representing each pair of margins:

$$R = \begin{bmatrix} 1 & \theta_{12} & \theta_{13} \\ - & 1 & \theta_{23} \\ - & - & 1 \end{bmatrix}.$$

For the GEE, θ represents the correlation parameter, while for the copula-based model, θ is the copula parameter for that pair of margins. For the copula-based model here we assume normal dependence for each pair-copula, as this is the implementation currently available in GJRM. For both the GEE and copula-based models, we specify the dependence structure as unstructured so as not to advantage the models with the additional information of the known AR-1 dependence structure in model selection.

2.2 Software

Initially developed for normal response distributions, random effects for intercepts and covariate slopes have been incorporated in multiple distributional regression frameworks including generalized additive models, (GAM [Hastie and Tibshirani, 1990](#)), and generalized additive models for location, scale and shape, (GAMLSS [Stasinopoulos et al., 2024](#)), making them a very accessible tool for modeling correlated data. GAMLSS provides a flexible regression framework that extends GAMs to allow mul-

multiple parameters of a response distribution to be modeled simultaneously. In addition, the range of potential response distributions is extended beyond the exponential family to any distribution with computable derivatives. Generalized estimating equations (GEE) have also been extended to apply to a much broader range of scenarios.

For GJRM and GLM fits, we use the **R** packages **GJRM** (Marra and Radice, 2023) and **glm** (R Core Team, 2024) respectively. For GEE fits, we use **glmttoolbox** (Vane-gas et al., 2023), which provides a modern implementation of the GEE with significant additional features for diagnostics and fitting.

For GLMMs, there are multiple packages available for fitting. In these simulations, we utilize four packages that use different methods. We include methods primarily based on their popularity and ability to capture the various different approaches used for the optimization of the fit. This enables us to compare not just GLMM to alternative models but also compare the implementations of the GLMM, as the methods for model fitting differ significantly. The methods for GLMM that are included are:

- The package **gamlss** (Rigby and Stasinopoulos, 2005) is included for fits of four and five parameters with normally distributed random effects. It utilizes the `random()` function to fit random effects and optimizes based on a penalized likelihood function;
- **lme4** (Bates et al., 2015) is included as it is one of the most popular packages for fitting generalized linear mixed models with random effects, and uses a Laplace approximation to the likelihood function;
- The function `gamm()` from **mgcv** (Wood, 2001) is included for fits of four parameters with normally distributed random effects. This utilizes `glmmPQL()` to optimize

a penalized quasi-likelihood function;

- **gamlss.mx** (Stasinopoulos and Rigby, 2024), in the **gamlss** suite of packages, is included for the fitting of a non-parametric random effect term in a model with five parameters.

We refer to these methods in shorthand as GAMLSS (4), GAMLSS (5), LME4, GAMM, and GAMLSS NP, respectively. Throughout our simulations, we found that the results from the GAMLSS (5) model were highly unstable, providing extreme results and failing to converge at a high rate. These have been excluded from the presented results.

All code used for this paper is available in the v1.0.0 release public repository at <https://github.com/ahibbert/bivariate-copula-for-correlated-data/releases/tag/v1>.

3 Results

Our evaluation of model performance covers three areas to provide a broad view of the difference in model fits across the methods presented, and to provide the ability to highlight areas of strength or weakness for the different models.

First, we evaluate estimated model **coefficients** and **standard errors** against the true values. This provides a view as to the models' suitability for inference, both in terms of the estimate and the standard error that would be used in hypothesis testing. Models which provide closer coefficient estimates to the true value on average, or provide more consistently close estimates are preferred. Similarly, providing accurate standard errors for parameters on average or more consistently close estimates to the

true values are preferred.

Second, we evaluate overall fit on the basis of **model selection criteria**. We use multiple criteria due to the advantages and drawbacks of each approach.

One criterion used is Bayesian Information Criteria (BIC), defined as $p \ln(n) - 2\ell$, where ℓ is the logarithmic likelihood of the model fit, p is the number of parameters and n is the sample size. BIC generally provides the more parsimonious model for inference compared to AIC, which provides the best model for prediction (Stasinopoulos et al., 2024). For reference we also present the log-likelihood in the main text, and AIC and GAIC (4) (AIC with a penalty for p of 4) in Appendix D. Some adjustment to the likelihood criteria is needed: for the GLMM models, conditional AIC (Donohue et al., 2011) (cAIC) and its extension to BIC has been used to take into account the variability of random effects on the degrees of freedom of the model. In the GAMLSS GLMM optimization method, the degrees of freedom are estimated by the degrees of freedom of the smoother needed to fit the random effect. This effective degrees of freedom (EDF) is calculated directly by the GAMLSS packages and is roughly comparable to the same estimate for EDF for a GLMM in the calculation of conditional AIC (cAIC) used for the AIC and BIC estimates for the LME4 models. The GAMLSS, GAMLSS NP and LME4 methods all provide valid likelihoods and so are adjusted to cAIC. Unfortunately, while this measure is highly sensitive, it is not available for the GEE as this uses quasi-likelihood for optimization; or GAMM as its optimization is based on penalized quasi-likelihood. Hence, we require an additional method to enable comparison of all models. The use of a method that does not score on the training data would be ideal to protect against overfitted models. We also note that while quasi-likelihood criteria are available for the GEE, they are

only comparable between GEE models.

The other criterion for model selection used is the variogram score ([Gneiting and Raftery, 2007](#)) with $p = 2$. This is an extension to the concept of the energy score, which provides significantly more sensitivity to incorrectly specified correlations, of interest for this model comparison. Variogram scores provide a pairwise comparison of all datapoints between a simulated fitted model and simulated realizations of the true dataset, essentially comparing the true distribution to that proposed by the model. The value of p is the power of the differences, so $p = 2$ provides squared pairwise differences. As the variogram Score does not rely solely on the realized dataset being fit, the score provides a criterion which is more robust against overfitted models than likelihood-based criteria. Variogram scores are shown to be sensitive to incorrectly predicted means, variances, and correlations ([Scheuerer and Hamill, 2015](#)), allowing us to assess the full distributional fit in a similar way to the log-likelihood. More information on the variogram score is given in [Appendix D](#).

Cross-validated Mean Squared Error of Prediction and Log Score were both considered as alternative approaches for assessment of overall fit. However, we lack a prediction method for the GJRM which can incorporate the effect of correlation for a time holdout, which would be required for an accurate out of sample assessment. We also lack the ability to predict for an unobserved individual in many models, which would be needed for an in-time-point holdout. Hence these methods are unable to provide a comparison between all methods and are not included.

Third, we evaluate model **computational complexity** through runtime and convergence of each model across the range of simulations.

Across the simulations, we use Kendall's τ to describe the strength of dependence between the random variables for the response at both time points, Y_1 and Y_2 . Kendall's τ represents the degree of concordance between two random variables, with a value in $(-1, 1)$ (Gijbels et al., 2011). As a measure of dependence, it has the advantage that it is not dependent on the relative scale of observations for continuous distributions, as is the case with Pearson correlation. We use the marginal skewness to provide additional information on the shape of the distribution, calculated as the average of the skewnesses of Y_1 and Y_2 .

3.1 Coefficient estimates and standard errors

Each of the models specified in Section 2.1 is applied to the simulated datasets. Estimates for parameters and their standard errors are compared to their true values under the simulated bivariate distribution.

The key results for the simulations are shown in Figure 1 for the intercepts and Figure 3 for the covariate coefficients β_{x_1}, β_{x_2} . As the bias and variance of the estimates appear to be highly dependent on Kendall's τ , we present these as a function of τ .

Importantly, please note that while we often refer to estimator bias throughout this section, for the GLMMs the targeted coefficient in their estimation is the coefficient conditional on the random effect. This means wherever we refer to bias for GLMM implementations throughout this section and related sections we are referring to bias in GLMM estimates if they are used for the purpose of estimating or interpreting true marginal coefficients, which is due to an estimand mismatch. This point is discussed in more detail in Section 3.1.3.

3.1.1 *Intercept coefficients*

Bias and standard error results for intercepts for times 1 and 2 are extremely similar, so for simplicity only results for time 1 are shown; results for time 2 are available in Appendix E. Figure 1 provides the average bias (left plots) and standard error (right plots) for time 1 intercept, for each of the models, across Kendall's τ of the simulated distributions. The black lines indicate the true asymptotic values for the bias (horizontal line at zero) and standard error. The true values of the parameters and their asymptotic standard errors are calculated based on simulation of the joint distributions in Appendix A, and the asymptotic properties of the maximum likelihood estimators.

Beginning with the **bivariate normal** (top row), the results show that all models exhibit minimal bias throughout the range of simulated distributions, except for the GJRM with an ill-fitting skewed copula (Clayton), which is included for comparison. The GJRMs, GEE and GLM provide close estimates for the standard error (SE); however, while LME4 and GAMM provide broadly accurate standard errors, the GAMLSS (4) model shows an incorrectly decreasing standard error for realizations with higher τ .

In contrast to the normal model, for the **non-normal models**: the bivariate negative binomial (second row), bivariate Gamma (third row) and bivariate Bernoulli (bottom row), we find substantial bias in the parameter estimates for the intercept from the parametric GLMMs (LME4, GAMLSS(4), GAMM) while other models are relatively unbiased. In investigating the characteristics of simulated distributions that result in biased parametric GLMM estimates, our finding is that higher values of Kendall's τ and / or higher values of skewness of the marginal distributions of the simulated bi-

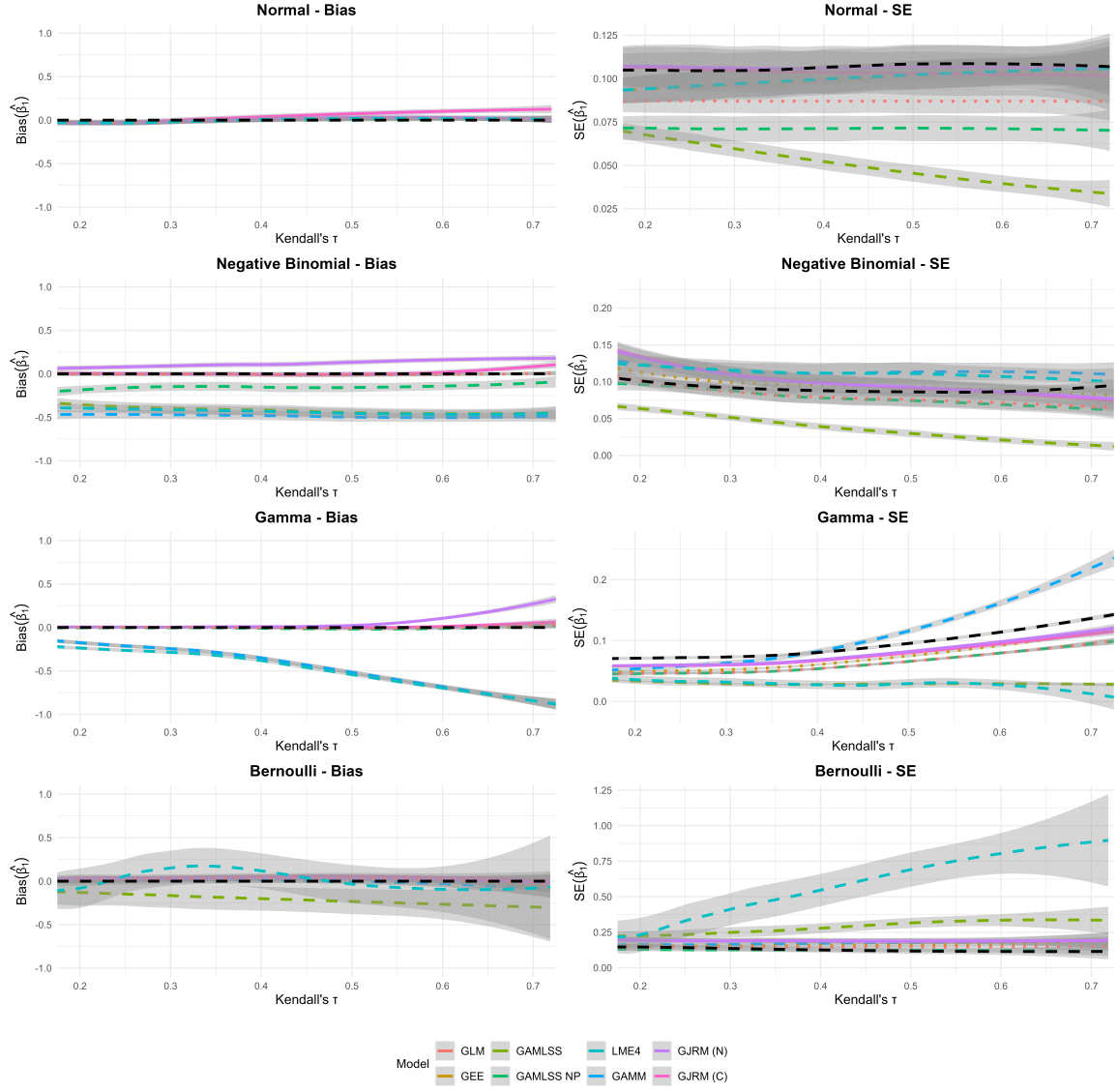


Figure 1: Bias for the estimate of the intercept for time 1 plotted with inverse link transform $g^{-1}(\beta_1)$ (left) and standard error for β_1 (right), across the range of realizations of the simulated distributions. Estimates are plotted against Kendall's τ which has been computed for each simulation. Random effect models (GLMMs) are shown as dashed lines, GJRMs are solid and GLM, GEE are dotted lines for ease of reference.

variate distribution result in higher bias. In addition, there are some large differences in estimates for parameters and standard errors between GLMM implementations.

For the **negative binomial**, bias for all four GLMM estimates (GAMLSS (4), LME4, GAMLSS NP (5), GAMM) is exhibited reasonably consistently across the range of Kendall's τ values, with the remaining models being relatively unbiased. For this distribution, skewness is a stronger indicator of GLMM bias. The bias against the marginal skew is provided in Figure 2. Standard error estimates for the negative binomial parameters are close to the true values for the GJRM, GEE, and GLM. The GLMMs differ substantially for estimates of the standard error, with the GAMLSS (4) model underestimating SE for higher τ , and LME4 and GAMM slightly overestimating SE across the range of τ .

In the case of the **bivariate Gamma**, bias for the parametric GLMMs increases almost linearly with increasing τ for the GAMLSS (4) model, GAMM and LME4. All other models provide unbiased estimates for the intercept except for the GJRM with a normal copula that is biased for higher τ , due to its ill-fitting copula. In extreme cases with rank correlation greater than 0.6, the GLMM estimates from the parametric GLMMs are less than half the simulated intercept. Interestingly, we also found that the LME4 and GAMLSS (4) parametric GLMMs provide much more inconsistent estimates for the intercept in the simulation range, with a much greater variation in estimate bias across values of τ compared to the other models. In comparison, the GLM, with a similar underlying structure but without adjustment for dependence, is robust, providing unbiased estimates for the intercept.

In terms of standard error for the Gamma, the GLM, GEE and GJRMs closely follow the true standard error. In contrast, the GLMM implementations again exhibit

differing behavior. The GAMLSS (4) and LME4 models provide a generally flat standard error with increasing values of τ , severely underestimating the true standard error, and the GAMM model follows the trend of the true standard error, but is overestimating the true standard error substantially for high τ .

Model bias for the intercept plotted against skewness for all non-normal models is shown in Figure 2. The GLMMs provide increasingly biased estimates for higher values of marginal skewness, whereas the remaining models remain relatively unbiased throughout the range.

Across the set of realizations, we find that for the Gamma, increasing marginal skewness holding τ constant, or increasing τ and holding skewness constant, results in increasing bias for the GLMMs. This is shown in Table 2 (LHS) for the non-time-variant- σ random effect model, GAMLSS (4). This is also the case for the GJRM with less-well fitting symmetric normal copula being fit to the skewed dependence of the bivariate Gamma (Table 2 (RHS)).

For the **bivariate Bernoulli** using a logit link (logistic model), the results are similar to those of the negative binomial and Gamma models. The GJRM, GEE and GLM all provide reasonable estimates of the marginal intercepts and their standard errors. However, for GLMMs, bias is present for marginal intercept estimates in the range of τ with higher bias for higher τ . However, while the GAMM estimates are biased, its estimates are much closer to the true values than the other parametric GLMMs, being only 5 percent different from the true value at the worst point compared to 20-30 percent for the other GLMMs. Again, for standard error, the GLMMs differ substantially, with the LME4 model providing extremely high standard errors, compared to the GAMLSS (4) model which is only slightly higher than the known true

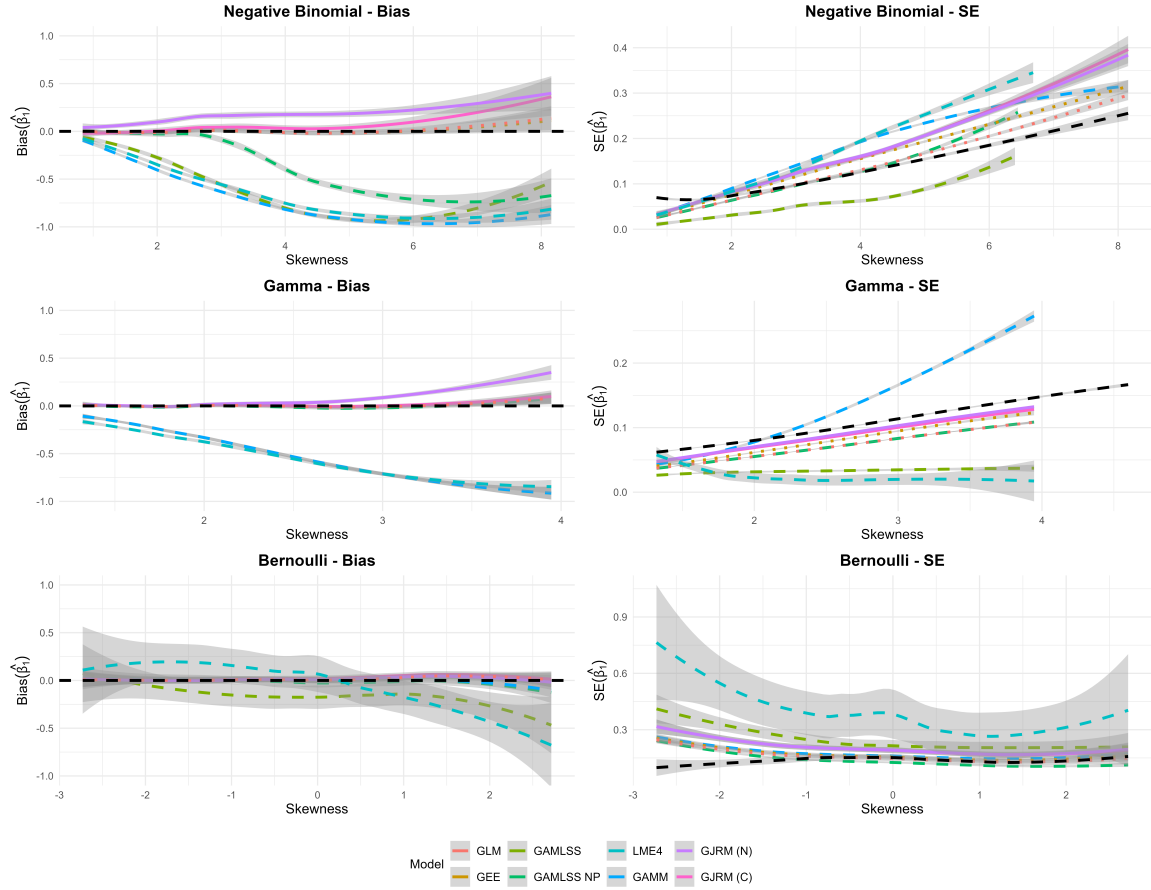


Figure 2: Bias for the estimate of the intercept for time 1 plotted with inverse link transform $g^{-1}(\beta_1)$ (left) across the range of realizations of the simulated distributions, plotted against average marginal skewness. Random effect models (GLMMs) are shown as dashed lines, GJRM are solid and GLM, GEE are dotted lines for ease of reference.

values, while the GAMM provides an accurate estimate in line with the other models and the true value.

In summary, for the bivariate normal, all models provide reasonable estimates of the intercept and its standard error for the specified parameters, except for the GJRM where the copula is ill-fitting. For the negative binomial, Gamma, and Bernoulli, the

Marginal skewness for Y_1					Marginal skewness for Y_1				
τ	1-2	2-3	3-4	4-5	τ	1-2	2-3	3-4	4-5
0.0-0.2	-0.099	-0.262	-	-	0.0-0.2	-0.001	-0.006	-	-
0.2-0.4	-0.233	-0.424	-0.722	-0.942	0.2-0.4	-0.024	-0.005	0.102	0.269
0.4-0.6	-0.318	-0.541	-0.822	-0.961	0.4-0.6	0.002	-0.016	0.105	0.304
0.6-0.8	-	-0.692	-0.853	-0.972	0.6-0.8	-	0.023	0.215	0.782
0.8-1.0	-	-	-	-0.976	0.8-1.0	-	-	-	1.050
GAMLSS (4)					GJRM (N)				

Table 2: Average bias for the estimate of β_1 presented in inverse link transform $g^{-1}(\beta_1)$ against τ and marginal skewness for realizations of the bivariate Gamma. Left hand side table is for the GAMLSS (4) model and right hand side table is for the GJRM with normal copula, which is ill-fitting for this distribution compared to the Clayton copula.

GLM and GEE provide consistently unbiased estimates of the intercept and standard error. The GJRM similarly provides unbiased estimates for the intercept and standard error across all distributions as long as the copula function is well specified. When the copula function is not well fitting, bias presents for high τ and high marginal skewness, similar to the GLMMs. For the GLMMs, the results differ substantially between packages. In general, we find that for the non-identity link models, the GLMMs provide increasingly biased estimates of the intercept with higher values of τ and marginal skewness. The GLMMs also often significantly under- or over-estimate the standard error with different trends depending on the package used, and inconsistency is higher with higher τ and marginal skew.

3.1.2 Coefficients β_{x_1} , β_{x_2}

Results for coefficients β_{x_1} and β_{x_2} are very similar so we only include results for β_{x_1} in Figure 3 and include β_{x_2} in Appendix E for reference. For the normal, all models are reasonably accurate for β_{x_1} and its standard error, except for GAMLSS which underestimates SE for high correlation, as was the case for β_1 .

Interestingly, for the negative binomial, the GJRM with ill-fitting copula provides the most biased estimates, particularly for high correlation, while the GJRM with well-fitting copula, and all other models, estimates are reasonable. However, the GAMLSS model also underestimates SE across the range for the negative binomial, while other GLMMs are reasonably accurate, though slightly higher than the true value and other models.

For the Gamma, all models are reasonably unbiased for β_{x_1} except for regions of very high correlation. At above $\tau = 0.6$, almost all models except the GJRM with normal copula underestimate the parameter by about 5 percent. SE is accurately captured by the GJRMs, GLM and GEE but is highly underestimated by the LME4 and GAMLSS models, while being highly overestimated by the GAMM. The only substantial difference for the estimation of β_{x_2} compared to that of β_{x_1} is that for the Gamma for high τ , the GLMMs overestimate β_{x_2} , as opposed to for β_{x_1} where they underestimate the coefficient.

Estimates for β_{x_1} for the Bernoulli are highly unusual. The GAMLSS and LME4 models provide highly biased over-estimates for β_{x_1} , but strangely the two models differ on SE estimates: the GAMLSS model heavily under-estimates the SE and the LME4 model heavily over-estimates it.

In summary, all models provide relatively reasonable estimates for coefficients for the normal case. For the other distributions, compared to the intercept estimates, the bias for coefficient estimates from the GLMMs is substantially reduced for β_{x_1} and β_{x_2} , but still present for high correlation and skewness for the negative binomial and Gamma, and still highly present for all Bernoulli models. In general, the SE results for β_{x_1} and β_{x_2} are inconsistent between the GLMM packages. The **gamlss** `random()` implementation consistently underestimates the SE of the coefficient to the extent that it would substantially change interpretation; **lme4** overestimates SE substantially for the Bernoulli and underestimates substantially for the Gamma; and **mgcv** `gamm()` software overestimates for the Gamma and is reasonable for the Bernoulli.

3.1.3 Reviewing the differences in coefficient estimates between GLMM and other models

The significant difference in bias for the GLMMs between the normal distribution models and the negative binomial and Gamma distribution models provides an indicator of the mechanism that drives the bias. There are three key differences between the normal and other distribution models: (i) most obviously, the marginal distributions; (ii) the shape of the dependence structure; and (iii) the link function. The results of simulations in this paper indicate that if using a non-identity link function with a distribution with significant skewness and/or correlation, then inferences based on conditional parameters may be wildly different from their marginal values. Essentially, the higher the correlation (measured by Kendall's τ) and the higher the marginal skewness, the greater distortion the random effect term will cause to the conditional distribution in comparison to the true marginal distribution.

Although unintuitive, this bias is expected behavior for the GLMM coefficient esti-

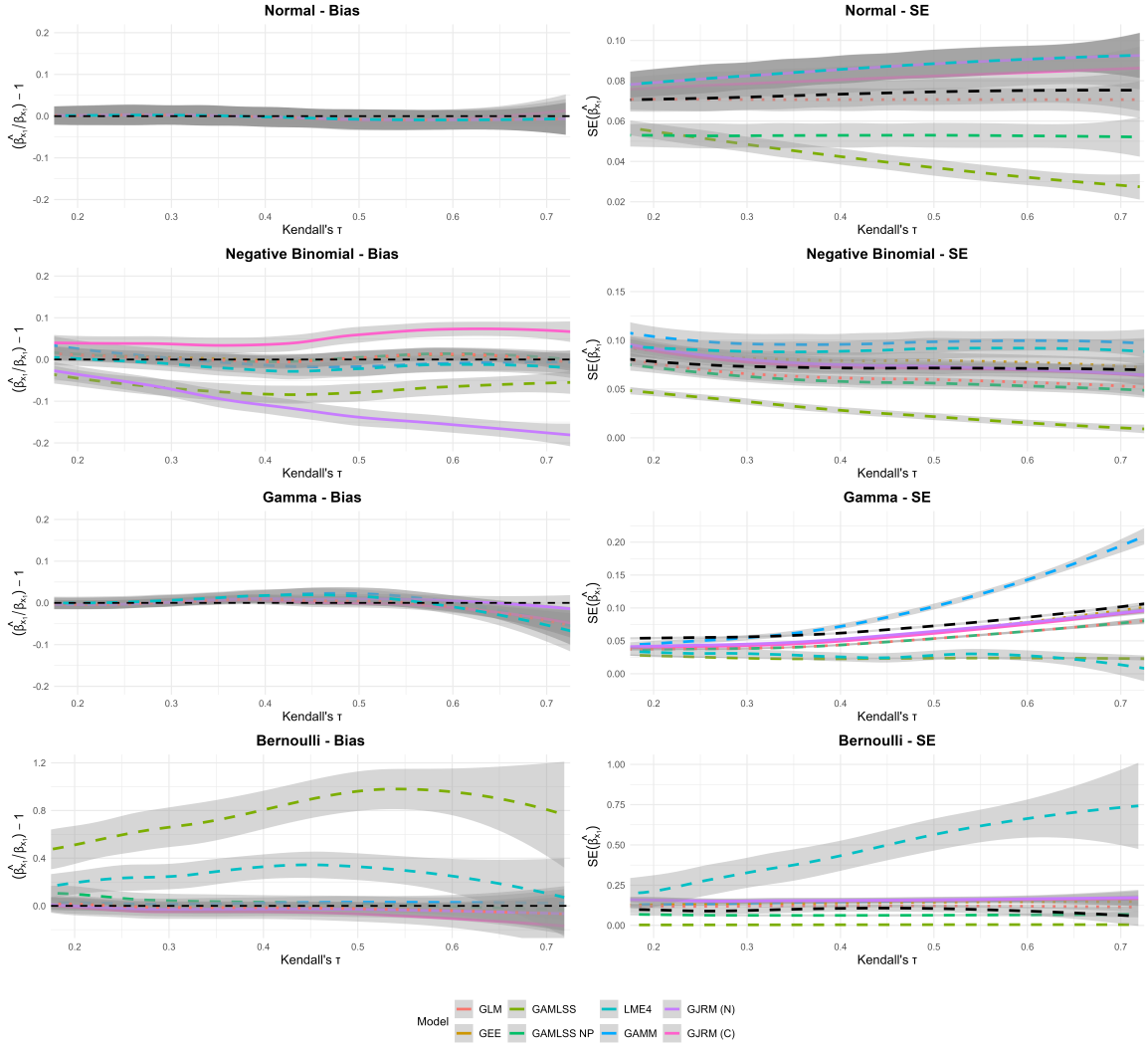


Figure 3: Bias for the estimate of the coefficient for x_1 across the range of realizations of the simulated distributions, plotted against Kendall's τ . Random effect models (GLMMs) are shown as dashed lines, GJRM as solid and GLM, GEE are dotted lines for ease of reference.

mates to some extent based on the maximum likelihood estimators due to the estimand mismatch. To illustrate the impact of the link function on the estimator, we introduce a new parameter $\beta_t = \beta_2 - \beta_1$. In Appendix C we show derivations for the exponential family, for the maximum likelihood estimators for β_1 , β_2 and β_t with the

log link, for a marginal and conditional model. We use these estimators to show why bias arises for parameters β_1 and β_2 but not for β_t . For simplicity we don't include covariates x_1 and x_2 in this comparison but the working extends to these.

For the parameters β_1 and β_2 we have the following maximum likelihood estimators.

For the marginal model:

$$\hat{\beta}_1 = \log(\bar{y}_1); \quad \hat{\beta}_2 = \log(\bar{y}_2)$$

For the conditional model:

$$\hat{\beta}_1 = \log(\bar{y}_1) - \log\left(\frac{1}{n} \sum_{i=1}^n e^{b_i}\right); \quad \hat{\beta}_2 = \log(\bar{y}_2) - \log\left(\frac{1}{n} \sum_{i=1}^n e^{b_i}\right).$$

So, the value for $\hat{\beta}_1$ and $\hat{\beta}_2$ will differ between conditional GLMMs and marginal models based on the scale of $\sum e^{b_i}$ relative to $\sum y_{it}$.

Interestingly, for the logarithmic link function, while the intercept estimates cannot be easily extracted from the conditional model as $E(Y_t) \neq E(Y_t|b)$, the model can be respecified so that we can extract one of the parameters without a difference between the conditional and unconditional (marginal) model estimates. Consider the model:

$$\log(\mu_{it}) = \beta_1 + \beta_t \mathbb{1}_{t=2} + b_i \tag{3.1}$$

In Appendix C we show that the MLEs for β_t for both marginal and conditional models are:

$$\hat{\beta}_t = \log(\bar{y}_2) - \log(\bar{y}_1).$$

So, the conditional and marginal estimates for β_t using the log link are equivalent, due to the cancellation of $\sum b_i$ terms. As confirmation, we have rerun all the models

in previous sections with the model respecified as Equation (3.1). We find that the results for the first parameter β_1 are consistent with the results presented for the GLMM in previous sections (biased and incorrectly estimating SE), while the time parameter β_t is unbiased (although still presents similar inconsistencies in standard error estimates as for other parameters from GLMM fits).

This specific simplification for the estimation of the time parameter, β_t in this model, is the only parameter we have shown to be unbiased for non-normal data with logarithmic link and is only valid for a longitudinal factor such as the effect of time in this model. For example, covariates that are constant over time, such as age at the beginning of the study or sex (e.g. x_1 and x_2), the factor would not be able to be extracted independently of b_i and will not provide unbiased estimates. We are not aware of other forms of parameter formulation that may be defined without bias presenting.

More generally, only a covariate that is specified such that its value is independent of b_i when using a given non-identity link function can be extracted without bias from the GLMM for a marginal parameter value. This form will differ depending on the link function.

3.2 Model selection criteria

The averages of the model selection criteria against τ are presented for all simulations. The results are presented as averages across parameter configurations that share the same values of τ , with equal weight provided to each simulation. Aggregation is performed using a fitted loess curve for each model.

The logarithmic likelihood is shown in Figure 4, the Bayesian Information Criteria (BIC) (Stasinopoulos et al., 2024) in Figure 5, and the unweighted and weighted variogram scores ($p = 2$) in Figure 6 and Figure 7. Note that we present $-2 \times \log$ -likelihood instead of the likelihood so that the three plots can be interpreted as the best model having the lowest result. The results for AIC and generalized AIC with penalty $k = 4$ (denoted GAIC (4)) are provided in Appendix D. The results for AIC and GAIC(4) are broadly comparable between GJRM and GAMLSS and so are not presented in detail.

The GEE does not provide a valid log-likelihood and, therefore, is not included in likelihood-based comparisons. This is a significant advantage of the GJRM method over the GEE, as it can be compared to alternative models such as the GLM and GLMM using standard likelihood-based criteria. For the variogram score, we are able to simulate all models except the non-parametric random effect model, GAMLSS NP.

GAMM provided identical log-likelihood results to LME4 for the normal, but provided extremely unusual likelihood results in comparison to other GLMMs for non-normal models, for example, five times lower log-likelihood for the logistic model, and so are excluded from comparison and require further investigation.

3.2.1 Likelihood-based model selection criteria

The findings across all models are quite consistent for likelihood-based criteria. The GAMLSS random effect model with 4 parameters (green dashed line) is consistently chosen as having the lowest (best) $-2 \times \log$ -likelihood across all distributions, while also having one of the two highest (worst) BIC alongside the LME4 model. The GJRMs provide the next-best log-likelihoods across most distributions after the GAMLSS

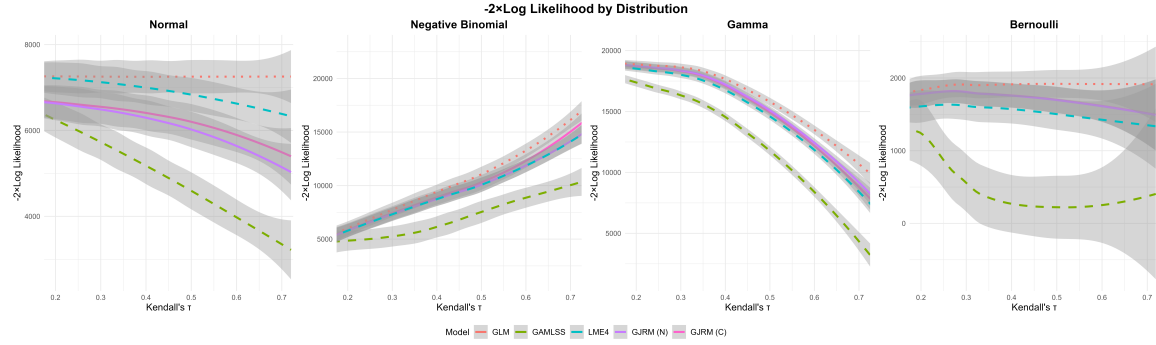


Figure 4: Average model selection criteria ($-2 \times \log$ -likelihood) for the likelihood-based models across the range of realizations of the simulated distributions. Estimates are plotted against Kendall's τ which has been computed for each simulation. Random effect models (GLMMs) are shown as dashed lines, GJRM are solid and GLM, GEE are dotted lines, for ease of reference.

model, while providing the lowest (best) BIC on average across all bivariate distributions. For the negative binomial and Gamma, the LME4 and GJRM models have comparable likelihood, while the GJRM is preferred by likelihood for the normal and the LME4 model is preferred by likelihood for the Bernoulli model.

The GLM and GAMLSS GLMM with a non-parametric random effect are rarely preferred to the GJRM or parametric GLMMs. Similarly, the LME4 model is rarely preferred to the GAMLSS GLMM. In terms of copula choice for the GJRM, as would be expected, the GJRM with normal copula is preferred for the multivariate normal and the GJRM with Clayton copula is preferred for the skewed dependence structure of the bivariate Gamma though only marginally, while being very similar for the remaining models. Broadly, both copulas seem to provide similar fit.

The high BIC results for the GLMMs is due to the high effective degrees of freedom (EDF) needed to account for the variability of the random effect used for those mod-

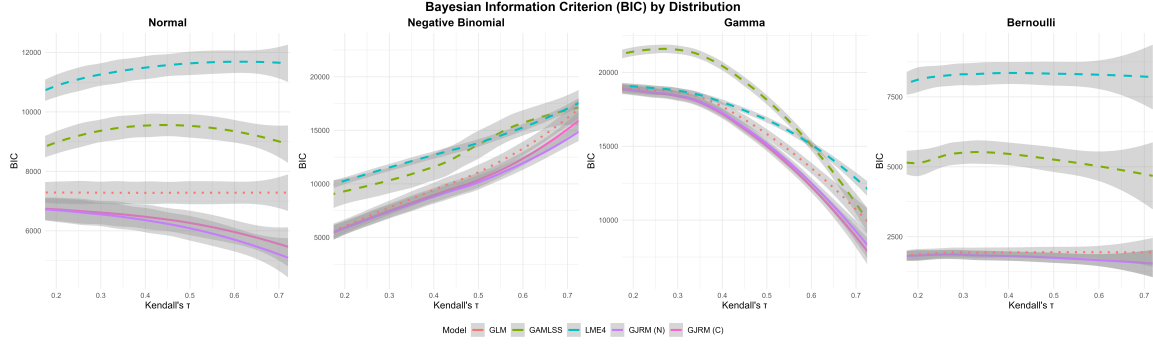


Figure 5: Average model selection criteria (BIC) for the likelihood-based models across the range of realizations of the simulated distributions. Estimates are plotted against Kendall's τ which has been computed for each simulation.

els. This means that model choice using standard model selection criteria between GAMLSS (4) and the GJRM is largely dependent on the penalty for the degrees of freedom of the random effect. The choice is between the more parsimonious model of the GJRM selected by BIC or the more closely fitting GLMM selected by log-likelihood.

3.2.2 Variogram-based model selection criteria

In addition to likelihood-based model selection criteria, we also compare model fit using variogram scores with $p = 2$, taking the variogram score over 100 simulations for each model fit for each distribution and combination of parameters.

Our initial findings were that the variogram score provided minimal differentiation between models for the Normal and Bernoulli, failing to differentiate even the GLM compared to correlation adjusting models at a high correlation. This is due to the nature of the bivariate data, in which correlated observations only account for $1/N$ of the total score, where N is the total number of observations across all time points.

In addition, correlated observations contribute less to the overall differences in the complete pairwise matrix used to calculate the variogram score due to their greater similarity than uncorrelated observations. To account for this problem in the score, we apply a weighting to the variogram calculation, which assigns a different weight to the subset of correlated observations in the dataset while setting the weight for all other observations to the default of 1. We find that a weighting of $4 \times N$ for the correlated observations provides a reasonable balance between assessing the marginal fit and the correlation structure. This weighting is proportional to the ratio of the number of total observation pairs to the number of correlated observation pairs combined with the average of the level of Pearson correlation in the variogram calculation matrix. Variogram scores with non-negative weightings are shown to be proper (but not strictly proper), in plain terms meaning the score is maximized when the true distribution is forecast ([Scheuerer and Hamill, 2015](#)). Variogram scores by definition are not strictly proper because a variogram score with parameter p cannot differentiate between distributions with moments which are the same up to p , but different for $p + 1$ or higher. Additional details regarding the weighting approach is included in [Appendix D](#).

The effect of the selected variogram weighting approach can be seen well in comparing the variogram scores in the first plot for the bivariate normal and the last plot for the bivariate Bernoulli in [Figure 6](#) (unweighted) and [Figure 7](#) (weighted). The unweighted scores provide almost identical scores with large overlapping ranges for all models, whereas weighted scores clearly differentiate the GLM (orange dotted line) as a worse fit (higher variogram score) with increasing correlation of the modeled dataset. While weighting provides additional differentiation for the normal and Bernoulli, the rankings for weighted and unweighted variogram scores are very similar

for the negative binomial and Gamma.

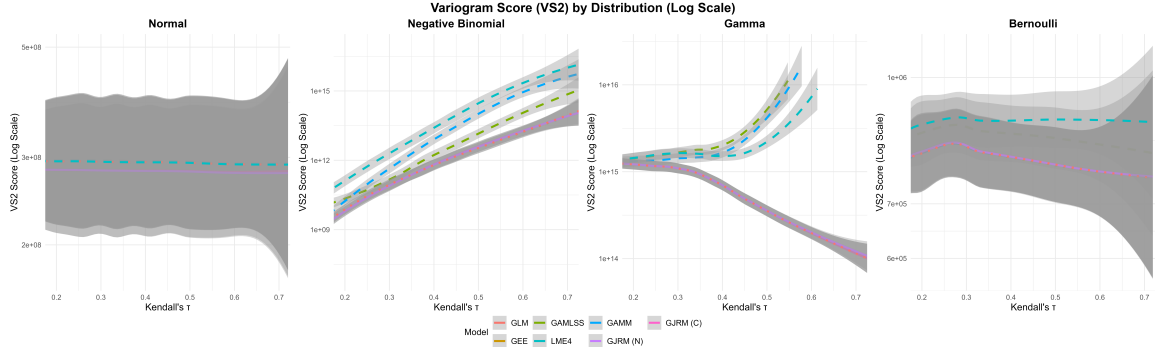


Figure 6: Average model selection criteria (log of unweighted variogram score) for the seven simulatable models (excluding non-parametric random effect GAMLSS NP) across the range of realizations of the simulated distributions. Estimates are plotted against Kendall's τ which has been computed for each simulation.

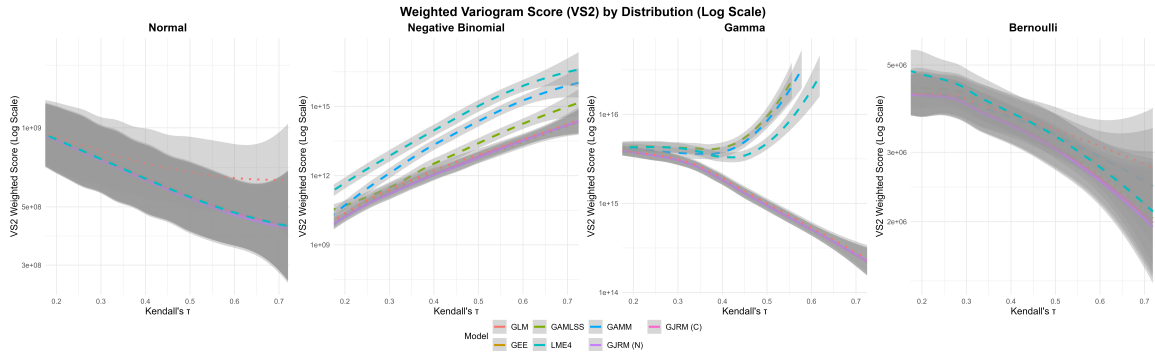


Figure 7: Average model selection criteria (log of weighted variogram score) for the models we can simulate (excluding non-parametric random effect GAMLSS NP) across the range of realizations of the distributions. Estimates are plotted against Kendall's τ which has been computed for each simulation.

In terms of results, for the normal and Bernoulli, with weighting applied the variogram scores consistently rank the GLM as the worst fitting model for high correlation. However, the scores do not differentiate between the fit of the remaining models with high variation in ranking between simulations.

For the negative binomial and Gamma, both of which use the log link, we find that the random-effect-based models (dashed lines) all provide increasingly worse fits compared to the other models with increasing correlation of the underlying dataset. A review of these situations finds that the issue is with the increasing contribution of the simulated random effect to the overall fit and its interaction with the link function. With larger random effect variance relative to the marginal model, simulated results in the tail of the random effect distribution are exponentiated and result in more highly skewed simulated distributions in comparison to the true marginal distribution. This may be the same issue that contributes to convergence issues in fitting GLMMs to these highly skewed datasets. In contrast, for the logistic model, this issue does not occur in simulations of the random effects, as the link function controls the impact of the random effect on the marginal distribution more effectively. Overall for the negative binomial and Gamma, the GEE and GJRM provide the best average variogram scores, both weighted and unweighted. The scores do not substantially differentiate between the fit of the GEE compared to the GJRM.

Surprisingly, while the weighted variogram score is able to differentiate the GLMM models when they provide extreme values for the log link distributions, and the GLM models which are ignoring correlation, the score does not seem to provide substantial differentiation for the more subtle differences between models which are adjusting for correlation.

3.3 Computational complexity

The computational complexity of each method is a significant potential factor influencing choice of model. With increasing complexity in model structure such as the

GLMM and GJRM over the GLM, comes increased time to compute.

To compare models, we provide the runtime results for each of the models over 100 simulations of two different parameter combinations at different ends of the correlation range, and at different sample sizes. The results are presented in Table 3. We find no significant differences in runtime for different parameter settings of the same distribution for a given model, so we simply provide timing results by distribution and sample size. For simplicity, we only present results for two of the GJRM models, with normal and Clayton copulas, since runtimes between copulas are extremely similar.

At a sample size of $n = 2000$, the order of model computation speed is generally consistent. In all cases, the GLM provides the fastest computation, followed by the GEE taking roughly ten times longer than the comparable GLM. Of the remaining models, the GJRMs are the fastest, averaging two seconds per run, substantially faster than the 4 seconds for GAMLSS NP, 6 seconds for GAMM and GAMLSS, and 9 seconds for LME4. The one exception is that for the normal, LME4 seems to be highly optimized and is in fact faster than the GEE just for this case. Conversely, LME4 has by far the longest runtime for the negative binomial models, and in fact is the only model which presents any convergence errors throughout these simulations. At $n = 2000$, the LME4 model provided convergence warnings for 70 percent of negative binomial fits and 97 percent of Gamma fits even with an extremely high iteration limit. At $n = 200$, LME4 provided much more stable results with only 1 percent of the models failing for the negative binomial and 69 percent for the Gamma. In contrast, the GAMLSS and GAMM models did not provide any convergence warnings for these fits, though it may be possible that the unusually low standard error results for the model across simulations may be caused by convergence errors which are not

being identified in the optimization approach.

We provide a ratio of the total runtime between the fits at $n = 200$ and $n = 2000$ to provide an indication of how the model runtime scales with the sample size. A model that scales linearly with the sample size would provide a ratio of 10x for this case. We find that the GLM provides the most favorable scaling with sample size at 3.6x, followed by the GJRM, which average 5.3x, followed by the remaining models, which are all above 7x. The GAMM and GAMLSS NP models in particular seem to scale the most unfavorably with this change in sample size.

Although these total run-time values are not particularly impactful for an individual model, while building and comparing multiple models, and as sample sizes increase in the era of big data, these computational differences become increasingly more noticeable.

These operations were performed on Fedora Linux 41 with an AMD Ryzen 7 7840U with 64 GB of RAM. The following R packages were used for estimation: **glm** (R Core Team, 2024), **glmtoolbox** (Vanegas et al., 2023), **gamlss** (Rigby and Stasinopoulos, 2005), **lme4** (Bates et al., 2015), **gamlss.mx** (Stasinopoulos and Rigby, 2024), **mgcv** (Wood, 2001) and **GJRM** (Marra and Radice, 2023) in R version 4.4.2.

3.4 Trivariate extension

For trivariate data, GJRM software only provides joint regression capability for Bernoulli response distributions of those reviewed in bivariate simulations. (Our reasons for restricting to bivariate and trivariate fits supported by GJRM are included in Section 1.1.) We extend simulations and fits from bivariate simulations to trivariate

Sample	Runtime	Models							
		GLM	GEE	GAMLSS	GAMLSS NP	LME4	GAMM	GJRM (C)	GJRM (N)
n=200	Gamma	0.006	0.04	0.42	0.34	0.36	0.12	0.21	0.14
	Bernoulli	0.006	0.03	0.49	0.16	0.22	0.09	0.09	0.03
	Normal	0.005	0.01	0.08	0.37	0.04	0.04	0.12	0.05
	Neg Bin	0.017	0.06	1.98	0.46	4.32	0.57	1.33	1.01
	Average	0.008	0.03	0.74	0.33	1.24	0.20	0.44	0.31
n=2000	Gamma	0.02	0.23	2.56	4.20	3.48	5.32	1.00	0.79
	Bernoulli	0.01	0.33	2.31	1.03	1.48	5.46	0.22	0.09
	Normal	0.01	0.14	0.42	3.60	0.09	5.16	0.32	0.19
	Neg Bin	0.07	0.45	16.57	7.12	30.95	10.04	6.71	6.12
	Average	0.03	0.29	5.47	3.99	9.00	6.49	2.06	1.80
Ratio of n=2000 / n=200	Gamma	3.2	6.3	6.1	12.4	9.7	46.2	4.8	5.8
	Bernoulli	1.8	10.9	4.8	6.4	6.7	57.6	2.6	3.4
	Normal	2.0	11.2	5.0	9.8	2.1	132.8	2.8	3.5
	Neg Bin	4.2	8.0	8.4	15.6	7.2	17.8	5.0	6.1
	Average	3.6	8.6	7.4	12.0	7.3	31.9	4.7	5.9

Table 3: Seconds runtime comparison for each of the fitted models for 100 simulations of two different parameter combinations of each simulated distribution. Runtime is presented for sample size of n=200 and n=2000. A ratio of the total runtime between the sample sizes is included to provide an indication of model runtime scaling with sample size. Linear scaling in runtime would be a 10x ratio.

Bernoulli distributions, for which joint regression software is available.

In general, we show that all findings for bivariate Bernoulli regression extend to trivariate Bernoulli models:

- **Coefficients:** The estimands for both intercepts and covariates from the GLMMs show increasing mismatch to the true marginal values with increasing values of correlation; while the GEE, GLM and GJRM provide accurate estimates and standard errors.
- **Model selection:** Variogram scores are unable to reasonably differentiate between models except for the extreme case of correlation 0.75 and above, where the LME4 model appears to provide extreme distributional fits. Weighted variogram scores select against GAMM and GLM for increasing correlation, as was the case for bivariate fits, with all other fits being reasonably similar. Log likelihood overwhelmingly selects GAMLSS followed by GJRM and LME4, while BIC overwhelmingly selects GJRM.

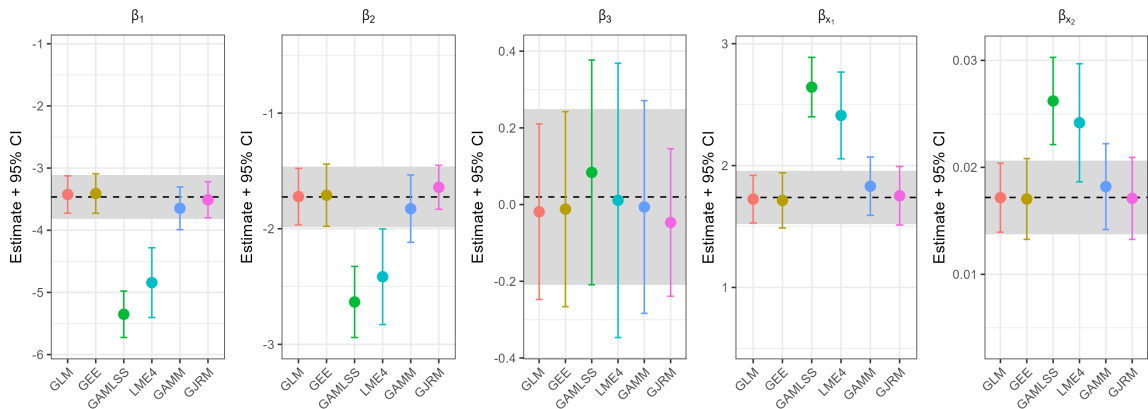


Figure 8: Trivariate Bernoulli; $\rho = 0.5$: fitted coefficients and 95 percent confidence intervals based on model estimates averaged over 1000 simulations compared to the asymptotic coefficient (black dotted line) and confidence interval.

For the trivariate Bernoulli, plots are provided for the average coefficient estimates and confidence intervals based on 1000 simulations. For the sake of brevity we show results for moderate correlation ($\rho = 0.5$, Figure 8). This level of correlation shows differences between models clearly, without the extreme results from GLMMs at higher correlations substantially skewing the charts. Additional charts for $\rho = 0.1$, $\rho = 0.25$, $\rho = 0.75$ and $\rho = 0.9$ are provided in Appendix F. As was the case for $T = 2$, the estimates from the GAMLSS and LME4 GLMMs are highly biased for both marginal intercepts and covariates due to estimator mismatch, with the bias being larger with greater correlation. Also as was the case for $T = 2$, the GAMM model significantly outperforms the other GLMMs in terms of coefficient estimate accuracy for the margin, though is generally still not quite as accurate as the GEE, GJRM and GLM. The GEE and GJRM both provide reasonably accurate estimates for all parameters and standard errors. Surprisingly, the GLM also provides strikingly accurate estimates and standard errors, almost matching the GEE and GJRM. In general, the coefficients and standard errors appear to be the most correct for the GEE, followed by the GJRM then GLM, though the differences are very small.

Results for model selection criteria over the same 1000 simulations are provided for moderate ($\rho = 0.5$, Figure 9) and high ($\rho = 0.75$, Figure 10) correlation with the remaining cases provided in Appendix F. As was the case for bivariate models, the variogram scores provide minimal differentiation between models. Interestingly, for high correlation the LME4 model fit in particular begins to provide extreme fits with high variogram scores compared to the alternative models, with other GLMM software remaining robust. Weighted variogram scores show the same trends as for bivariate models; GAMM and GLM are differentiated as worse fits with higher correlation while the remaining models are broadly comparable. In terms of likelihood-based criteria,

GAMLSS is generally selected by log-likelihood and AIC for any ρ , followed by the GJRM. The GJRM is always selected by BIC.

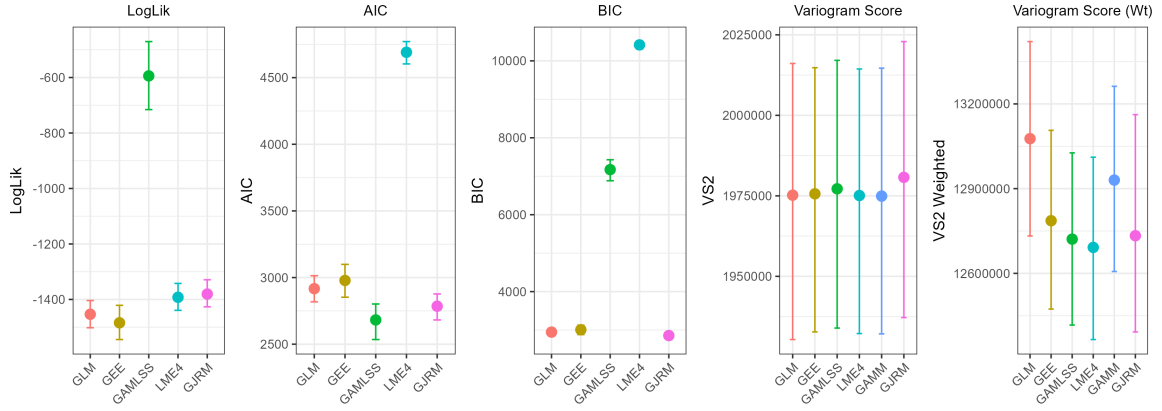


Figure 9: Trivariate Bernoulli; $\rho = 0.5$: model selection criteria based on model estimates over 1000 simulations of variogram scores from 100 realisations of each model. GEE quasi-likelihood included for reference.

4 Application

We introduce a dataset with distributional properties similar to those described in the simulations. In this setting, compared to the simulation setting, the marginal distributions are not known in advance, so their parameters must be estimated and fit to be able to analyze the dependence structure. The shape of the dependence structure is then interrogated, and a copula of best fit is selected.

4.1 Doctor visits in RAND Health and Retirement Longitudinal Study

The dataset considered is from the RAND Health and Retirement Study ([RAND Center for the Study of Aging, 2023a](#)) ([RAND Center for the Study of Aging, 2023b](#)),

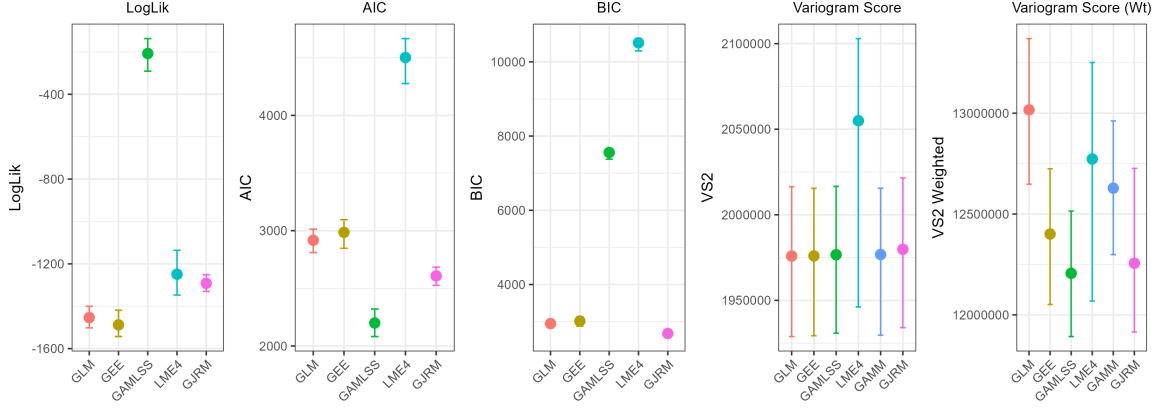


Figure 10: Trivariate Bernoulli; $\rho = 0.75$: model selection criteria based on model estimates over 1000 simulations of variogram scores from 100 realisations of each model. GEE quasi-likelihood included for reference.

which is a large longitudinal study of different aspects related to population aging in the United States. The HRS (Health and Retirement Study) is sponsored by the National Institute on Aging (grant number NIA U01AG009740) and is conducted by the University of Michigan. An outcome measure of interest is the number of visits to a doctor during the two years between surveys. We choose the two most recent survey time points as an example case, 2018 (time 1) and 2020/21 (time 2).

The sample used consists of 2,404 individuals with observations at both time points. The mean number of doctor visits at time 1 is 8.6 and at time 2 is 7.7. The distribution of number of visits is highly skewed with marginal skewness of 13.7 and 5.7 at times 1 and 2, respectively. Figure 11 shows the marginal distributions, which appear zero-inflated with modes at zero and around five. The best-fitting marginal distributions were interrogated using GAMLSS software and it was found that, using the AIC criterion, the zero-inflated Sichel (ZIS) distribution (Stasinopoulos et al., 2024) was the best fit for both marginal distributions. Marginal negative binomial and zero-

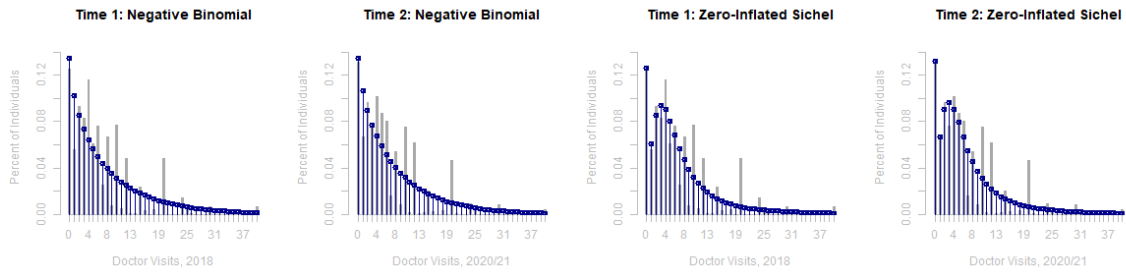


Figure 11: RAND Health and Aging study: Marginal distributions for time 1 and time 2 with fitted negative binomial distributions and fitted Zero Inflated Sichel distributions.

inflated Sichel distribution fits are shown fitted against the two margins. It can be seen that the zero-inflation is not accurately captured by the standard negative binomial. The number of visits is correlated, with a reasonably strong rank correlation measured by Kendall's τ of 0.40, and weak Pearson correlation of 0.28.

We perform a regression for the number of doctor visits. The same models specified in previous sections are fit, with covariates $x_1 = \text{sex}$ and $x_2 = \text{age}$. The results are shown for both a negative binomial distributional fit in Table 4, which is available to all modeling packages, and a ZIS fit using GAMLSS together with a separately estimated two-stage ZIS–Frank copula model in Table 5.

Table 4 shows the results for the models fit with a **negative binomial marginal distribution**. As seen in simulations, it is immediately clear that the parametric GLMMs (LME4, GAMLSS (4) and GAMM) provide significantly different estimates for the intercepts at times 1 and 2. In simulations, substantially differing estimates from GLMMs were generally biased due to estimand mismatch. Estimates for β_{x_1} are all comparable, but for β_{x_2} , all parametric GLMMs provide approximately twice the effect size of other comparable models, with confidence intervals which would strongly

exclude the alternative effect size. These suggest quite different models overall.

In terms of SE, surprisingly, even though the LME4 and GAMM models provide different parameter estimates, their SEs are quite similar to the other models that adjust for correlation. In contrast, the GAMLSS (4) and GAMLSS NP models provide substantially lower SE than the comparable models, suggesting greater confidence in the estimates. However, in simulations, standard errors for the GLMM for the negative binomial that were lower than the GLM for the coefficients were inconsistent with the true standard error.

In terms of model selection for the negative binomial models, the GAMLSS (4) model provides the best fit by AIC, however, by GAIC (4) and by BIC, the GJRM with the Frank copula provides the best fit. By AIC, the GJRM also provides the second best fit, outperforming LME4. The GEE also appears to provide a similar level of fit to the GJRM based solely on coefficient estimates and standard errors. Variogram scores select the GEE as the best model both weighted and unweighted, followed by the GJRMs (unweighted) and GLM (weighted), while selecting against the GLMMs as was the case for simulations. Note that variogram scores are not sensitive to moments beyond the order of p ; in this case, $p = 2$. Due to this limitation, comparisons between differently parameterized models should be viewed with caution given that the best fitting distribution for this data by likelihood-criteria has fits for both tail skewness and zero-inflation. Unfortunately, our review did not find that using $p > 2$ provided consistently different results; however, there may be alternative weighting patterns or transformations that would better account for this distributional shape that we are unaware of. There is some variability in rankings between simulations, so boxplots of the variogram scores for 100 realizations of the fitted models over 100 repeated

Model	Estimates				Standard Errors				Likelihood Criteria			Variogram		
	$\hat{\beta}_1$	$\hat{\beta}_2$	$\hat{\beta}_{x_1}$	$\hat{\beta}_{x_2}$	$\hat{\beta}_1$	$\hat{\beta}_2$	$\hat{\beta}_{x_1}$	$\hat{\beta}_{x_2}$	AIC	GAIC (4)	BIC	VS2	VS2 (w)	EDF
GLM	1.60	1.49	0.20	0.06	0.11	0.11	0.034	0.015	30239	30247	30244	.0814	.6608	5
GEE	1.60	1.49	0.20	0.06	0.16	0.17	0.056	0.023	-	-	-	.0802	.6604	6
GAMLSS (4)	0.77	0.68	0.18	0.11	0.06	0.06	0.018	0.008	27649	31117	30044	.0815	.6621	1734
GAMLSS NP	1.33	1.22	0.20	0.09	0.10	0.10	0.031	0.014	30031	30045	30041	-	-	7
LME4	0.70	0.61	0.20	0.13	0.14	0.14	0.044	0.020	30624	32117	31655	.0848	.6635	747
GAMM	0.59	0.50	0.20	0.14	0.14	0.14	0.045	0.020	-	-	-	.0813	.6614	-
GJRM (C)	1.65	1.93	0.17	0.03	0.15	0.16	0.039	0.018	29558	29576	29570	.0811	.6609	9
GJRM (N)	1.75	1.90	0.12	0.03	0.14	0.15	0.035	0.017	29587	29605	29600	.0813	.6616	9
GJRM (J)	1.59	1.73	0.18	0.05	0.15	0.15	0.036	0.017	29937	29955	29950	.0811	.6612	9
GJRM (G)	1.69	1.86	0.17	0.04	0.15	0.15	0.037	0.017	29732	29750	29744	.0809	.6614	9
GJRM (F)	1.62	1.89	0.16	0.04	0.15	0.15	0.041	0.019	29398	29416	29411	.0810	.6613	9

Table 4: RAND Health and Aging study: regression results for negative binomial models estimating number of doctor visits for individuals in the two years prior to survey date. Best two model selection results in each column are bolded for ease of reference. β_{x_2} is scaled per 10 units. Variogram score is presented in log scale. Identical digits for the variogram scores before the decimal place are removed.

simulations are provided in Appendix G, with just the average score provided in Tables 4 and 5. The ranking between the median and mean is identical, so only the mean is provided.

Table 5 presents fits with zero-inflated Sichel (ZIS) margins. ZIS is available in **gamlss** but is not currently implemented in **GJRM**. We therefore fit two ZIS margins using GAMLSS and subsequently fit a Frank copula using **VineCopula** (Czado and Nagler, 2022). This is a separately optimized two-stage ZIS-copula model, not a jointly estimated GJRM. Standard errors are obtained manually through simulation

from the fitted copula distribution followed by transformation through the fitted ZIS marginal CDFs. Because the margin and dependence components are estimated separately, the procedure may be less efficient than joint estimation ([Manner, 2007](#)), and its results may not be representative of the performance of a joint ZIS GJRM.

Across the board, the more flexible GAMLSS models with ZIS marginal distributions provide much better fit by AIC than the best fit exponential family distribution, the negative binomial, for the same data. The GAMLSS model with ZIS distribution has 1,316 lower AIC than the negative binomial GAMLSS fit, and the two-stage model with ZIS margins and Frank copula has 429 lower AIC than the best exponential family GJRM fit. This is unsurprising given the clear zero-inflation present in the dataset. In comparing the two ZIS models, GAMLSS and copula-based, we find that the GAMLSS ZIS model is preferred by AIC, while the copula-based ZIS model is preferred by GAIC (4) and BIC due to the high effective degrees of freedom for the GAMLSS ZIS model. The copula-based ZIS model is also preferred to the GAMLSS ZIS model by both weighted and unweighted variogram scores. Results from the ZIS models are on average not preferred to the NB alternative by variogram scores. However, both models provide a wide range of variogram scores, outperforming the NB alternatives in a portion of simulations. In general, variogram scores select the GEE over both ZIS models; this may be due to the previously mentioned limitations of the variogram score for models with distributional differences in higher moments.

Coefficient estimates for x_1 and x_2 are similar for the two ZIS models, however, importantly, the standard error estimates for the GAMLSS ZIS model are substantially lower than the two-stage ZIS–Frank model, as was the case in simulations. In simulations, low SE values from the GAMLSS model were generally inconsistent with the

other random effect models and were substantially lower than the true values. This poses a concern for interpretation of these values, particularly as they result in tighter confidence intervals. For example, $\hat{\beta}_{x_1}$ is extremely significant ($p < .00001$) based on the GAMLSS model but only just significant ($p = 0.02$) for the two-stage model.

Overall, the best-fitting models by likelihood-based criteria are the GAMLSS model and the copula-based models for both the negative binomial and zero-inflated Sichel distributional fits. Between the two models, model selection relies on the degrees of freedom penalty, and preference for interpretation. The copula-based models are selected as the best model by likelihood criteria for penalty of 4 or greater for degrees of freedom, otherwise the GAMLSS model is selected. The weighted and unweighted variogram scores with $p = 2$ consistently select the ZIS copula-based model over the ZIS GAMLSS random effect model fits. In addition, the GJRM model can be interpreted directly, while the GAMLSS estimates must be carefully interpreted based on the conditioned random effect term which is transformed by the log link. The GEE also appears to provide a highly competitive fit, providing the best variogram scores on average.

5 Limitations

The primary limitation of the study is the restriction to the comparison of longitudinal datasets in only two dimensions (or three dimensions for the Bernoulli). Ideally, performance of copula methods should be compared for any possible longitudinal model able to be fit by the GLMM or GEE, for example 10, 100 or 1,000 time points. There is some evidence that GLMMs utilizing penalized likelihood may perform poorly in the bivariate case (see Nelder in discussion of Rigby and Stasinopoulos ([Rigby and](#)

Model	Estimates				Standard Errors				Likelihood Criteria			Variogram		
	$\hat{\beta}_1$	$\hat{\beta}_2$	$\hat{\beta}_{x_1}$	$\hat{\beta}_{x_2}$	$\hat{\beta}_1$	$\hat{\beta}_2$	$\hat{\beta}_{x_1}$	$\hat{\beta}_{x_2}$	AIC	GAIC (4)	BIC	VS2	VS2 (w)	EDF
GAMLSS (ZIS)	0.80	0.72	0.18	0.12	0.06	0.06	0.02	0.01	26333	29079	35227	.0856	.6631	1373
Two-stage ZIS-Frank	1.17	1.45	0.16	0.10	0.13	0.13	0.07	0.03	28969	28991	29040	.0815	.6615	11

Table 5: RAND Health and Aging study: regression results for flexible GAMLSS-based models estimating number of doctor visits for individuals in the two years prior to survey date. Marginal distributions used are Zero-Inflated Sichel (ZIS). Best models by selection result are marked in bold. Variogram score is presented in log scale. Identical digits for the variogram scores before the decimal place are removed.

[Stasinopoulos, 2005](#))), as opposed to higher dimensions, which should be a priority for comparison when approaches for joint copula longitudinal regression in greater dimensions are developed. In addition, the advantage of the copula approach is more pronounced in more than two dimensions as it can model a time- or covariate-dependent strength of correlation.

The other limitation of the copula method is that it relies on complete data: every observation must be available for every observational unit. Naturally, this is often not the case for longitudinal studies where individuals may leave a study over time. There is an approach that accounts for censoring in a survival setting that has been developed ([Dettoni et al., 2020](#)), however this is not available more broadly and will be an essential feature for these types of models in more than two dimensions.

Other limitations that we have deliberately excluded to maintain a reasonable scope for the study include:

- Sample size: We did not find any evidence of substantial differences between models

by sample size in early investigations so did not include this dimension in further review.

- Broader range of covariate shapes, e.g. smooth terms: Considering the substantial differences in results between models for more simple factors such as intercept, binary and linear, we did not believe it essential to highlight further covariate fit differences. This could be reviewed in further analysis.
- Bayesian methods: There are Bayesian methods for both copula and alternative approaches to longitudinal regression, however, given this comparison already includes seven different models, we restrict our approach to only likelihood and quasi-likelihood approaches. Future reviews may incorporate Bayesian methods.

6 Conclusion

Simulations were run of two-timepoint longitudinal datasets generated from bivariate distributions with normal, Gamma, Bernoulli and negative binomial margins and varying non-standard dependence structures, as well as a three-timepoint Bernoulli. We compare fits from a copula-based Generalized Joint Regression Model (GJRM) with popular methods for modeling correlated datasets including Generalized Linear Mixed Models (GLMM) implemented in four different packages: **gamlss**, **lme4**, **mgcv**, and **gamlss.mx**, Generalized Estimating Equations (GEE) implemented in **glmttoolbox** and Generalized Linear Models (GLMs). We evaluated model performance on the basis of coefficient estimate accuracy, model selection criteria of variogram scores and Bayesian information criteria (BIC), and computational complexity.

We find that all models provide relatively accurate estimates of the coefficients for the bivariate normal distribution with identity link, and comparable fit by model

selection criteria, except for the GLM which does not account for correlation.

For non-normal bivariate distributions, the Gamma and negative binomial with logarithmic link models, and the bivariate and trivariate Bernoulli with logit link models, we find that the GLMMs with parametric random effects provide highly biased estimates for marginal intercept coefficients and covariate coefficients, due to estimand mismatch. In addition, we find that popular GLMM modeling packages **lme4**, **gamlss** and **mgcv** often provide inconsistent coefficient estimates and standard errors for the same model. In general, we find that bias due to estimand mismatch and inconsistencies for the GLMMs are more pronounced the higher the skewness of the marginal distributions and/or the rank correlation between the outcome variables. In contrast to the GLMMs, we find that the GJRM and GEE provide accurate estimates for model coefficients and their standard errors across all simulated distributions.

In terms of overall model fit, we find that the GJRM almost always outperforms the GLMMs and other likelihood-based models included when compared on the basis of BIC, while providing the second best log-likelihood after the GAMLSS GLMM implementation on average across all distributions. The quasi-likelihood-based GEE also provides competitive fits, with the GEE and GJRM providing a similar average variogram score (both weighted and unweighted) that is the lowest (best) of all included models on average. In contrast, the GLMMs provide some of the worst fits by this measure for the log link Gamma and negative binomial models. The GLMM fit become worse with high levels of rank correlation. In contrast, the GEE coefficient estimates and standard errors, as well as variogram score performance, suggest they provide a similar fit to the GJRM; however they lack the additional tools provided by a likelihood-based approach and the control over the correlation structure provided

by the GJRM fit.

We fit all models to a real-world longitudinal study of doctor visits and demonstrate that the GAMLSS and GJRM models provide the best overall fit for an exponential family distribution model by likelihood criteria. The GEE also provides comparable coefficient estimates to the GJRM and the best fit as measured by variogram scores. When the limitation of the exponential family is removed, the GJRM and GAMLSS models provide vastly superior models compared to the negative binomial alternative with significantly improved AIC, as they are able to capture the zero-inflation and skew present in the dataset more accurately.

7 Discussion

Mixed models are extremely popular as a relatively straightforward method of modeling dependent data, using the mechanism of a univariate framework combined with a stochastic effect that is common across dependent observations. More complex structures, such as hierarchical models, are readily implemented. However, the random effect is an artificial construct and is introduced for the purpose of inducing dependence, indirectly. Generalized joint regression models, on the other hand, use the copula as the correlation-inducing construct. This is an approach which imposes the dependence structure in a more direct and controllable way. Careful modeling of the nature of the dependence structure is made possible with the choice of copula and also the ability to model copula parameters with covariates. (We are not aware of GLMM capability to model the dependence structure with covariates.) GEE models fall somewhere between these two in that they offer a degree of control over the dependence structure while providing competitive fit performance and readily interpretable

coefficients. However, the drawback of not having a likelihood-based structure limits the tools available for analysis such as AIC/BIC and likelihood-based testing.

The results of this study indicate that the application of mixed models with random intercept terms to non-normal bivariate correlated data using standard packages without major adjustment provides misleading results for marginal coefficients due to estimand mismatch, and this is exacerbated when marginal distributions are skewed and/or rank correlation is high. In addition, given that our findings show deviation in results between GLMM implementations, we suggest it is essential to compare results from different packages if the GLMM is the chosen model. More generally, when using a non-identity link function for a random-intercept model for bivariate data, users should be extremely careful in parameter interpretation. Estimates should be thoroughly interrogated, particularly if marginal skewness or rank correlation are high.

Given the performance of GJRM in this study both in terms of parameter estimates and fit criteria, they provide a strong alternative to GLMMs and GEEs for modeling bivariate correlated data and trivariate Bernoulli data with a transparent approach, likelihood-based structure and simpler interpretation than the conditional structure of the GLMM.

The greater flexibility, higher level of transparency, and computational simplicity of generalized joint regression modeling pose significant opportunities for improving the methods used for longitudinal data analysis, once the methods are extended beyond the bivariate and trivariate Bernoulli cases.

Declaration of conflicting interests

The authors declared no potential conflicts of interest with respect to the research, authorship and/or publication of this article.

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Appendix

A Bivariate distributions

1.1 Bivariate normal

Much of the theory of random effect models and generalized estimating equations was developed around the multivariate normal distribution, which we use as a benchmark for the performance of the models. We specify the five-parameter bivariate normal model as:

$$\begin{pmatrix} Y_1 \\ Y_2 \end{pmatrix} \sim \mathcal{N}_2 \left(\begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \begin{pmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix} \right)$$

having $E(Y_t) = \mu_t$, $\text{Var}(Y_t) = \sigma_t^2$ and $\text{Corr}(Y_1, Y_2) = \rho$, for $t = 1, 2$.

We generate 225 realizations of the distribution with 1,000 observations at both time points with fixed marginal means of $\mu_1 = 1$ and $\mu_2 = 2$; varying values of σ_1 and σ_2 between 0.25 and 2.5; and correlation between 0.1 and 0.9. The **R** package **mvt-norm** ([Genz and Bretz, 2009](#)) is used for the generation of random numbers.

1.2 Bivariate negative binomial

We use a bivariate compound Poisson model ([Stein and Juritz, 1987](#)) to introduce a skew to our bivariate simulations, as well as a non-Gaussian dependence structure. This is constructed as

$$Y_1|\lambda, t_1 \sim P(\lambda t_1) \quad \text{independently of} \quad Y_2|\lambda, t_2 \sim P(\lambda t_2)$$

and

$$\lambda \sim \text{Gamma}(\mu, \sqrt{\sigma}),$$

where the above Gamma distribution uses the GAMLSS parameterization GA ([Stasinopoulos et al., 2024](#)), with moments:

$$E(\lambda) = \mu; \quad \text{Var}(\lambda) = \mu^2 \sigma.$$

The resultant distribution is a bivariate negative binomial:

$$\begin{pmatrix} Y_1 \\ Y_2 \end{pmatrix} \sim \text{BivNB}(\mu_1, \mu_2, \sigma)$$

where $\mu_t = \mu t$, for $t = 1, 2$. The marginal distributions are NBI(μ_t, σ) (GAMLSS parameterization of the negative binomial) with moments

$$E(Y_t) = \mu_t$$

$$\text{Var}(Y_i) = \mu_t + \sigma \mu_t^2 \quad \text{for } t = 1, 2,$$

$$\text{Cov}(Y_1, Y_2) = \mu_1 \mu_2 \sigma.$$

In this case λ acts as a non-normal random effect term that mixes in a multiplicative way, as opposed to the additive way a random intercept is introduced in the standard

mixed model. For this distribution, we generate 400 realizations with 1,000 observations at both time points for μ_1 and μ_2 between 5^{-3} and 5^3 and σ between 5^{-1} and 5.

1.3 Bivariate Gamma

We use the four-parameter bivariate Gamma of Nadarajah and Gupta ([Nadarajah and Gupta, 2006](#)), which is described in Appendix [B](#). This distribution, which we denote as $\text{BivGamma}(\mu_1, \mu_2, \sigma, \theta)$, allows us to introduce significantly more skewed continuous distributions, and includes a unique dependence shape, which cannot be directly described by a parametric copula, nor generated by random effect. We align with the GAMLSS parameterization of the Gamma distribution ([McCullagh and Nelder, 1989](#); [Stasinopoulos et al., 2024](#)):

$$\begin{pmatrix} Y_1 \\ Y_2 \end{pmatrix} \sim \text{BivGamma}(\mu_1, \mu_2, \sigma, \theta)$$

The marginal distributions are $\text{GA}(\mu_t, \sigma)$ with moments

$$\begin{aligned} \text{E}(Y_t) &= \mu_t \\ \text{Var}(Y_t) &= \sigma^2 \mu_t^2 \quad \text{for } t = 1, 2, \\ \text{Corr}(Y_1, Y_2) &= \frac{\sigma \sqrt{\theta}}{1 + \sigma^2 + \sigma^2 \theta}, \end{aligned}$$

where $\mu_t > 0$ and $\sigma > 0$ have the same meanings as in the GAMLSS parameterization GA ; and $\theta > 0$ is present only in the correlation.

The shapes of the bivariate Gamma distribution and its dependence structure, for selected parameter values, are illustrated in Figure [12](#). In Section [3](#), the distribution is simulated with $n = 1,000$ observations at both time points for 400 parameter

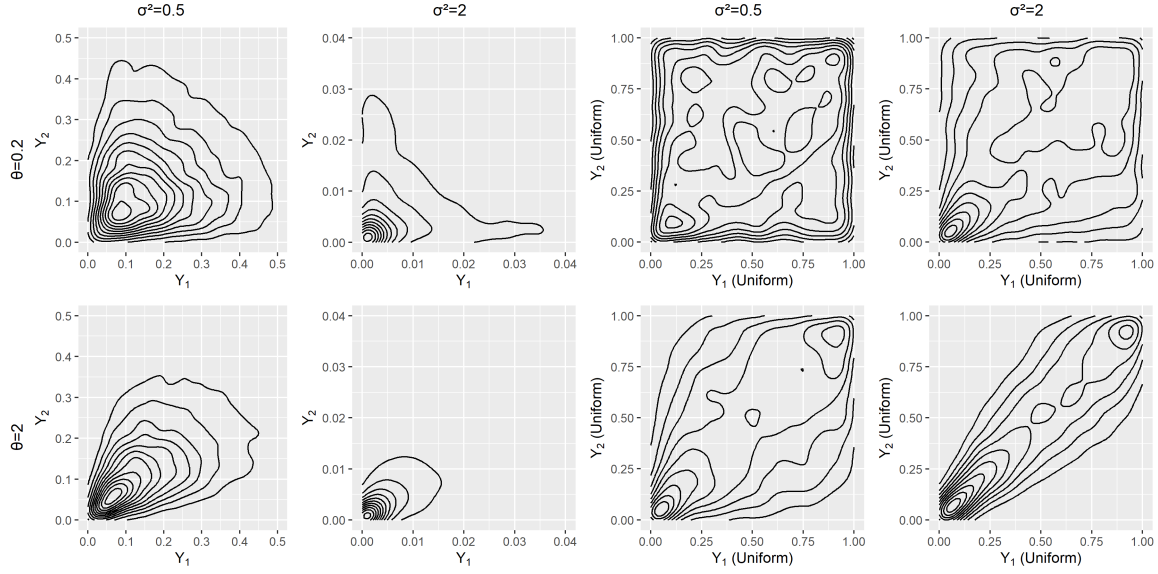


Figure 12: Shapes of the bivariate distribution of Nadarajah and Gupta for a simulation of 10,000 observations. Left four plots are contour plots of the bivariate distribution of (Y_1, Y_2) , for selected values of σ and θ , where $\mu_2 = 0.8\mu_1$ and $\mu_1 = 0.2$ when $\sigma^2 = 0.5$ and $\mu_1 = 0.05$ when $\sigma^2 = 2$. The right four charts are contour plots of the uniform transforms of the corresponding marginal distributions plotted against one another, to show the dependence structure with the marginal shapes removed.

combinations. The parameter settings chosen are: $\mu_1 \in (2, 3, \dots, 21)$; $\mu_2 = 1.2\mu_1$; $\sigma \in (\sqrt{0.2}, \sqrt{0.3}, \dots, \sqrt{2.1})$; $\theta \in (0.2, 0.3, \dots, 2.1)$. The resulting correlation varies between 0.08 and 0.4, and the rank correlation, as measured by Kendall's τ , between 0.04 and 0.82. The distribution exhibits a higher marginal skew with higher values of σ and a higher rank correlation with higher values of θ , with the highest rank correlation when σ is high and θ is high.

1.4 Bivariate Bernoulli

The logistic model is extremely common in life sciences and allows us to introduce another link function to simulations: the logit link. We firstly generate simulations from a bivariate standard normal:

$$\begin{pmatrix} Z_1 \\ Z_2 \end{pmatrix} \sim \mathcal{N}_2 \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix} \right)$$

Then we transform the margins such that $Y_t = \mathbb{1}(Z_t < C_t)$ where C_t is defined as a set of different cut-off points to represent a success or failure for each observation corresponding to selected values for the mean at both time points: $\mu_1 = \Phi(C_1)$ and $\mu_2 = \Phi(C_2)$. This results in the bivariate Bernoulli distribution with Bernoulli marginals and the following moments:

$$\mathbb{E}(Y_t) = \mu_t$$

$$\text{Var}(Y_t) = \mu_t(1 - \mu_t) \quad \text{for } t = 1, 2,$$

$$\text{Cov}(Y_1, Y_2) = \mathbb{P}(Z_1 < C_1, Z_2 < C_2) - \mu_1\mu_2$$

The parameter settings chosen are: μ_1 and μ_2 in $(.1, .25, .5, .75, .9)$ so multiple combinations of different class balance could be tested, and correlation ρ was tested in the range of values from 0.1 to 0.9.

1.5 Trivariate Bernoulli

We also simulate trivariate Bernoulli distributions. Our approach is to simulate a trivariate gaussian distribution with AR-1 dependence using **mvtnorm** and transform the margins to uniform, resulting in a multivariate normal dependence. We then impose a marginal Bernoulli distribution with binary covariate x_1 and uniform

covariate x_2 , with coefficients $\beta_{x_1} = 1$ and $\beta_{x_2} = 0.01$. As we find no substantial differences between differing intercept terms in bivariate simulations, we simply select a static set of intercepts common between simulations. We simulate for correlation between 0.1 and 0.9.

B Bivariate gamma distribution used for simulation studies

The bivariate Gamma used ([Nadarajah and Gupta, 2006](#)) induces dependence between two Gamma distributions by multiplying by a common Beta-distributed random variable, with some restriction on the choice of parameters. The approach is as follows:

1. Generate three independent random variables:

$$W \sim \text{Beta}(\alpha, \beta)$$

$$U \sim \text{Gamma}(\alpha + \beta, 1/\mu'_1)$$

$$V \sim \text{Gamma}(\alpha + \beta, 1/\mu'_2)$$

2. Let $Y_1 = WU$ and $Y_2 = WV$. Multiplication by W induces dependence between Y_1 and Y_2 .

The properties of the bivariate Gamma distribution of Y_1 and Y_2 , with parameters $\mu'_1 > 0, \mu'_2 > 0, \alpha > 0$ and $\beta > 0$ ([Yeo and Milne, 1991](#)) are as follows. The marginal distributions are:

$$Y_1 \sim \text{Gamma}(\alpha, 1/\mu'_1), \quad E(Y_1) = \alpha/\mu'_1$$

$$Y_2 \sim \text{Gamma}(\alpha, 1/\mu'_2), \quad E(Y_2) = \alpha/\mu'_2$$

with correlation coefficient

$$\text{corr}(Y_1, Y_2) = \rho = \frac{\sqrt{\alpha\beta}}{\alpha + \beta + 1}.$$

The parameters α and β fully define the dependence structure between the two marginal Gamma distributions. The bivariate distribution is described by its probability density function (PDF):

$$f(y_1, y_2) = C\Gamma(\beta)(y_1 y_2)^{\alpha+\beta-1} \left(\frac{y_1}{\mu_1} + \frac{y_2}{\mu_2} \right)^{\frac{\alpha-1}{2} - (\alpha+\beta)} \\ \exp \left[-\frac{1}{2} \left(\frac{y_1}{\mu_1} + \frac{y_2}{\mu_2} \right) \right] W_{\alpha+\frac{1-\alpha}{2}, \alpha+\beta-\frac{\alpha}{2}} \left(\frac{y_1}{\mu_1} + \frac{y_2}{\mu_2} \right),$$

for $y_1 > 0, y_2 > 0$ and where the constant C is given by

$$\frac{1}{C} = (\mu_1 \mu_2)^{\alpha+\beta} \Gamma(\alpha + \beta) \Gamma(\alpha) \Gamma(\beta)$$

and $W_{\lambda, \mu}$ is the Whittaker function ([Abramowitz and Stegun, 1972](#)):

$$W_{\lambda, \mu}(p) = \frac{p^{\mu+\frac{1}{2}} \exp(-p/2)}{\Gamma(\mu - \lambda + 1/2)} \int_0^\infty t^{\mu-\lambda-1/2} (1+t)^{\mu+\lambda-1/2} \exp(-pt) dt.$$

All the dependence structures from this bivariate Gamma incorporate a skew towards lower value dependence with differing strength of overall dependence, and a differing extent to which higher value dependence is also exhibited. The highest rank correlation within the dependence structure is achieved when α is low and β is high.

The second parameter of the multiplicative beta distribution, β , has no effect on the marginal distributions of Y_1 and Y_2 but is an important component in defining the dependence between the two variables. α and β together determine the strength of

the rank correlation of the dependence structure. α has the added effect of increasing marginal skewness with lower values.

We reparameterize this distribution to align with the GAMLSS parameterization of the Gamma distribution (Stasinopoulos et al., 2024). We define μ_t to capture marginal means, that is, $\mu_t = \alpha/\mu'_t$, and define $\sigma = 1/\sqrt{\alpha}$. We rename β to avoid confusion with the parameter used elsewhere in the text, and align with the notation used for the copula parameters, $\theta = \beta$. The bivariate Gamma in our notation is then

$$\begin{pmatrix} Y_1 \\ Y_2 \end{pmatrix} \sim \text{BivGamma}(\mu_1, \mu_2, \sigma, \theta)$$

where the marginal distributions are Gamma distributions in the GAMLSS (GA) parameterization:

$$Y_t \sim \text{GA}(\mu_t, \sigma), \quad \text{E}(Y_t) = \mu_t, \quad \text{Var}(Y_t) = \mu_t^2 \sigma^2 \quad \text{for } t = 1, 2$$

and

$$\text{corr}(Y_1, Y_2) = \frac{\sigma\sqrt{\theta}}{1 + \sigma^2 + \sigma^2\theta}.$$

C Maximum likelihood estimators for exponential family log link model parameters

For the exponential family we write the log-likelihood as:

$$\log [f(y_i|\mu, \phi)] = \frac{y_i \log(\mu) - \mu}{\phi} + \log [c(y_i, \phi)]$$

We investigate the maximum likelihood estimators for the GLM under independence, and GLMM conditional on random effects, and under two alternative parameterizations.

GLM, independence, original parameterization: For the **marginal model** with parameters μ_1 and μ_2 , where $\log(\mu_1) = \beta_1$ and $\log(\mu_2) = \beta_2$, we find the maximum likelihood estimators as follows:

$$\begin{aligned}\mathcal{L} &= \sum_{i=1}^n \log [f(y_i | \mu_1, \mu_2, \phi)] = \sum_{i=1}^n \frac{y_{i1}\beta_1 - e^{\beta_1} + y_{i2}\beta_2 - e^{\beta_2}}{\phi} + \log [c(y_i, \phi)] \\ \frac{\partial \mathcal{L}}{\partial \beta_1} &= \sum_{i=1}^n \frac{y_{i1} - e^{\beta_1}}{\phi} = 0 \implies \hat{\beta}_1 = \log(\bar{y}_1) \\ \frac{\partial \mathcal{L}}{\partial \beta_2} &= \sum_{i=1}^n \frac{y_{i2} - e^{\beta_2}}{\phi} = 0 \implies \hat{\beta}_2 = \log(\bar{y}_2)\end{aligned}$$

GLMM, original parameterization: For the **conditional model** with parameters μ_1 and μ_2 , where $\log(\mu_{i1}) = \beta_1 + b_i$ and $\log(\mu_{i2}) = \beta_2 + b_i$, we find the maximum likelihood estimators as follows:

$$\begin{aligned}\mathcal{L} &= \sum_{i=1}^n \log [f(y_i | \mu_1, \mu_2, \phi, b_i)] = \sum_{i=1}^n \frac{y_{i1}(\beta_1 + b_i) - e^{\beta_1 + b_i} + y_{i2}(\beta_2 + b_i) - e^{\beta_2 + b_i}}{\phi} + \log [c(y_i, \phi)] \\ \frac{\partial \mathcal{L}}{\partial \beta_1} &= \sum_{i=1}^n \frac{y_{i1} - e^{\beta_1 + b_i}}{\phi} = 0 \implies \hat{\beta}_1 = \log(\bar{y}_1) - \log\left(\frac{1}{n} \sum_{i=1}^n e^{b_i}\right) \\ \frac{\partial \mathcal{L}}{\partial \beta_2} &= \sum_{i=1}^n \frac{y_{i2} - e^{\beta_2 + b_i}}{\phi} = 0 \implies \hat{\beta}_2 = \log(\bar{y}_2) - \log\left(\frac{1}{n} \sum_{i=1}^n e^{b_i}\right)\end{aligned}$$

GLM, independence, reparameterized: For the **marginal model** with parameters β_1 and β_t where $\log(\mu_1) = \beta_1$ and $\log(\mu_2) = \beta_1 + \beta_t$:

$$\begin{aligned}\mathcal{L} &= \sum_{i=1}^n \log [f(y_i | \beta_1, \beta_t, \phi)] = \sum_{i=1}^n \frac{y_{i1}(\beta_1) - e^{\beta_1} + y_{i2}(\beta_1 + \beta_t) - e^{\beta_1 + \beta_t}}{\phi} + \log [c(y_i, \phi)] \\ \frac{\partial \mathcal{L}}{\partial \beta_t} &= \sum_{i=1}^n \frac{y_{i2} - e^{\beta_1 + \beta_t}}{\phi} = 0 \implies \hat{\beta}_1 + \hat{\beta}_t = \log(\bar{y}_2)\end{aligned}$$

substitute in value for $\hat{\beta}_1 = \log(\bar{y}_1)$

$$\hat{\beta}_t = \log(\bar{y}_2) - \log(\bar{y}_1)$$

GLMM, reparameterized: For the **conditional model** with parameters β_1 and

β_t where $\log(\mu_1) = \beta_1 + b_i$ and $\log(\mu_2) = \beta_1 + \beta_t + b_i$:

$$\begin{aligned}\mathcal{L} &= \sum_{i=1}^n \log [f(y_i | \beta_1, \beta_t, b_i, \phi, b_i)] \\ &= \sum_{i=1}^n \frac{y_{i1}(\beta_1 + b_i) - e^{\beta_1 + b_i} + y_{i2}(\beta_1 + \beta_t + b_i) - e^{\beta_1 + \beta_t + b_i}}{\phi} + \log [c(y_i, \phi)] \\ \frac{\partial \mathcal{L}}{\partial \beta_t} &= \sum_{i=1}^n \frac{y_{i2} - e^{\beta_1 + \beta_t + b_i}}{\phi} = 0 \implies \hat{\beta}_1 + \hat{\beta}_t = \log(\bar{y}_2) - \log\left(\frac{1}{n} \sum_{i=1}^n e^{b_i}\right) \\ &\text{substitute in value for } \hat{\beta}_1 = \log(\bar{y}_1) - \log\left(\frac{1}{n} \sum_{i=1}^n e^{b_i}\right) \\ \hat{\beta}_t &= \log(\bar{y}_2) - \log(\bar{y}_1)\end{aligned}$$

The above shows that the estimator for the conditional model for β_t does not depend on b_i and is equivalent to the estimator from the marginal model.

D Additional model selection comparison for models

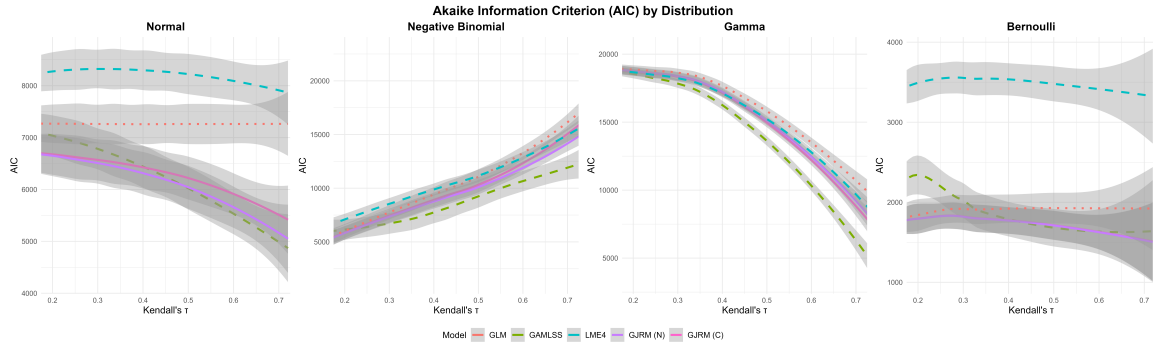


Figure 13: Average model selection criteria (AIC) for the likelihood-based models across the range of realizations of the simulated distributions. Estimates are plotted against Kendall's τ which has been computed for each simulation.

Regarding the derivation of the weight for the weighted variogram scores, our rationale is as below.

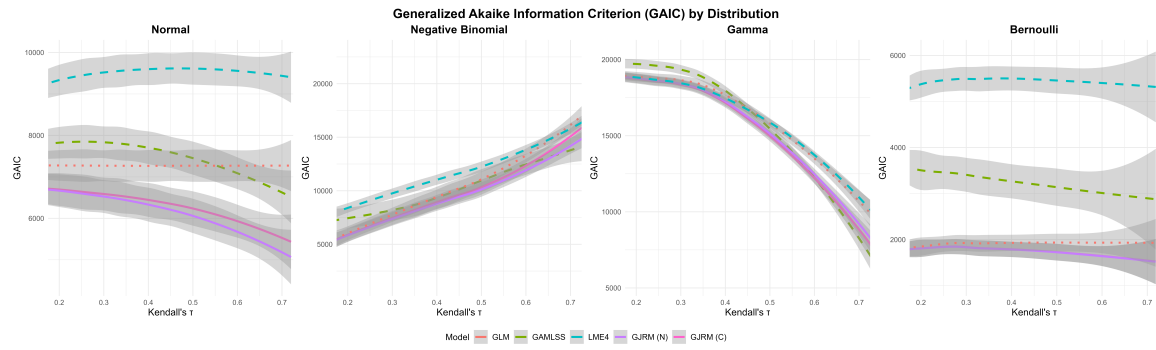


Figure 14: Average model selection criteria (GAIC (4)) for the likelihood-based models across the range of realizations of the simulated distributions. Estimates are plotted against Kendall's τ which has been computed for each simulation.

The variogram score with $p = 2$ is calculated based on squared pairwise differences between each pair of observations in the true distribution (or realizations from the known true distribution) compared to the squared pairwise differences between observations in the simulated models of the data. For a given longitudinal dataset with T time points and n observational units, with a total sample size $N = nT$, there are N^2 pairs, and the contribution of any given pair of observations which contribute to the variogram score is $1/N^2$. Given our interest is comparing correlation adjusting models, pairs which relate to correlated observations are of particular interest. These correlated observations only occur in the dataset when the pair is comprised of two observations of the same observational unit at a different time point. This results in a situation where if T is high relative to n then correlated observations comprise a higher proportion of the fit. For example, if $T = 2$ and $n = 2$ then we have 4 observations and 16 observation pairs of which 4 are correlated, or 25 percent. If $T = 2$ and $n = 1000$ as in our simulations, we have the extreme situation where we have 2,000 observations and 4 million observation pairs (N^2) of which only 2,000 (N) relate to correlated observations, or 0.05 percent. So to proportionally allow the correlation

structure to provide a similar weight to the overall score as the marginal shape, we suggest a weight inversely proportional to this ratio which is $N/N^2 = 1/N$. In addition, we want to adjust for the fact that any given correlated observations will count less to the overall score than uncorrelated observations due to the correlation causing smaller differences in both the true and simulated distributions. For simplicity we apply a flat increase proportional to an average level of 50 percent correlation squared rather than calculating correlation per distribution and account for the squared power difference, $0.5^2 = 0.25$, resulting in a final weighting of the inverse of $0.25/N$, $4N$.

E Coefficients additional results

F Trivariate additional results

G Applications additional results

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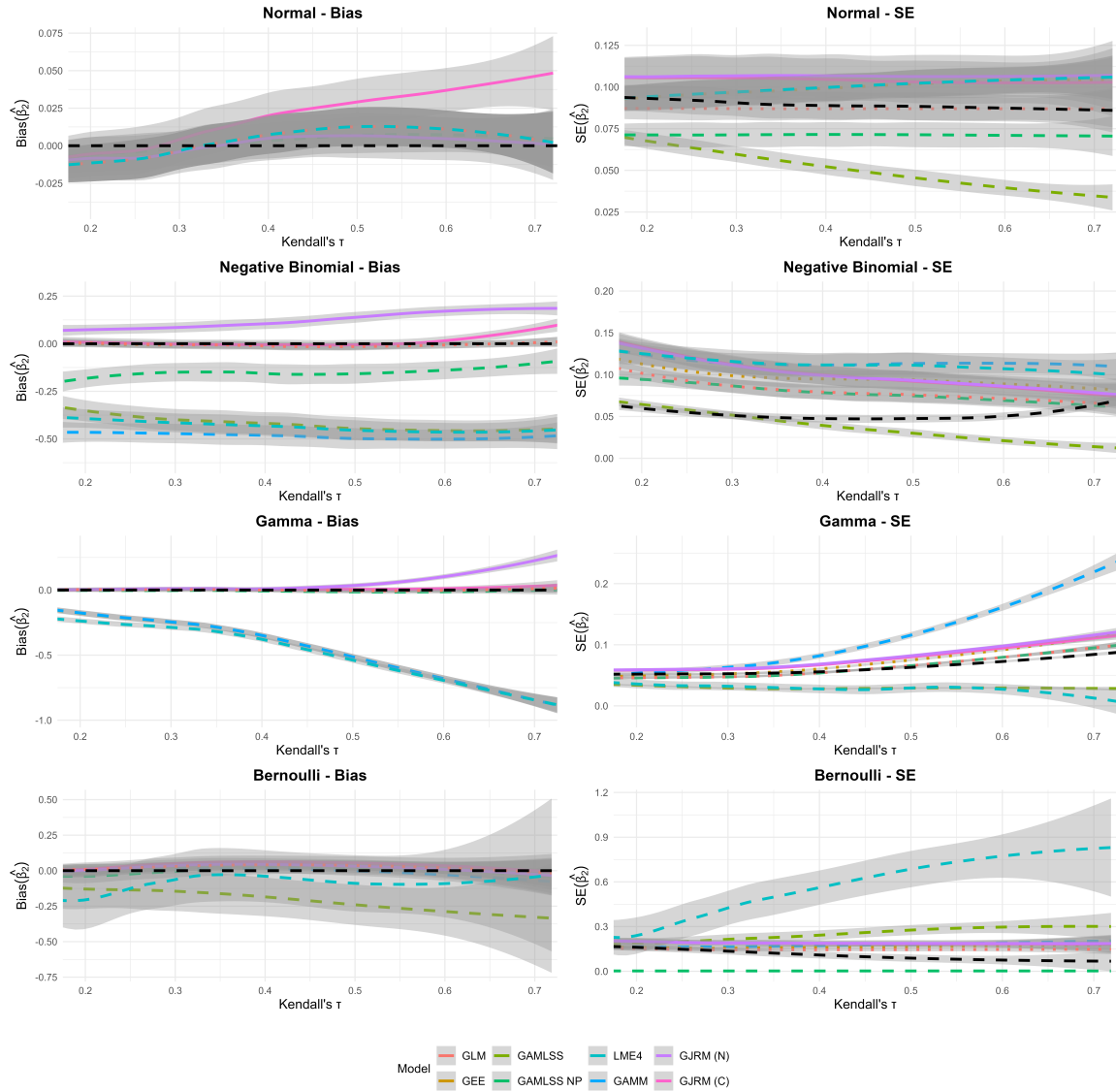


Figure 15: Bias for the estimate of the intercept for time 2 plotted with inverse link transform $g^{-1}(\beta_2)$ (left) and standard error for β_2 (right), across the range of realizations of the simulated distributions, plotted against Kendall's τ . Random effect models (GLMMs) are shown as dashed lines, GJRM are solid and GLM, GEE are dotted lines for ease of reference.

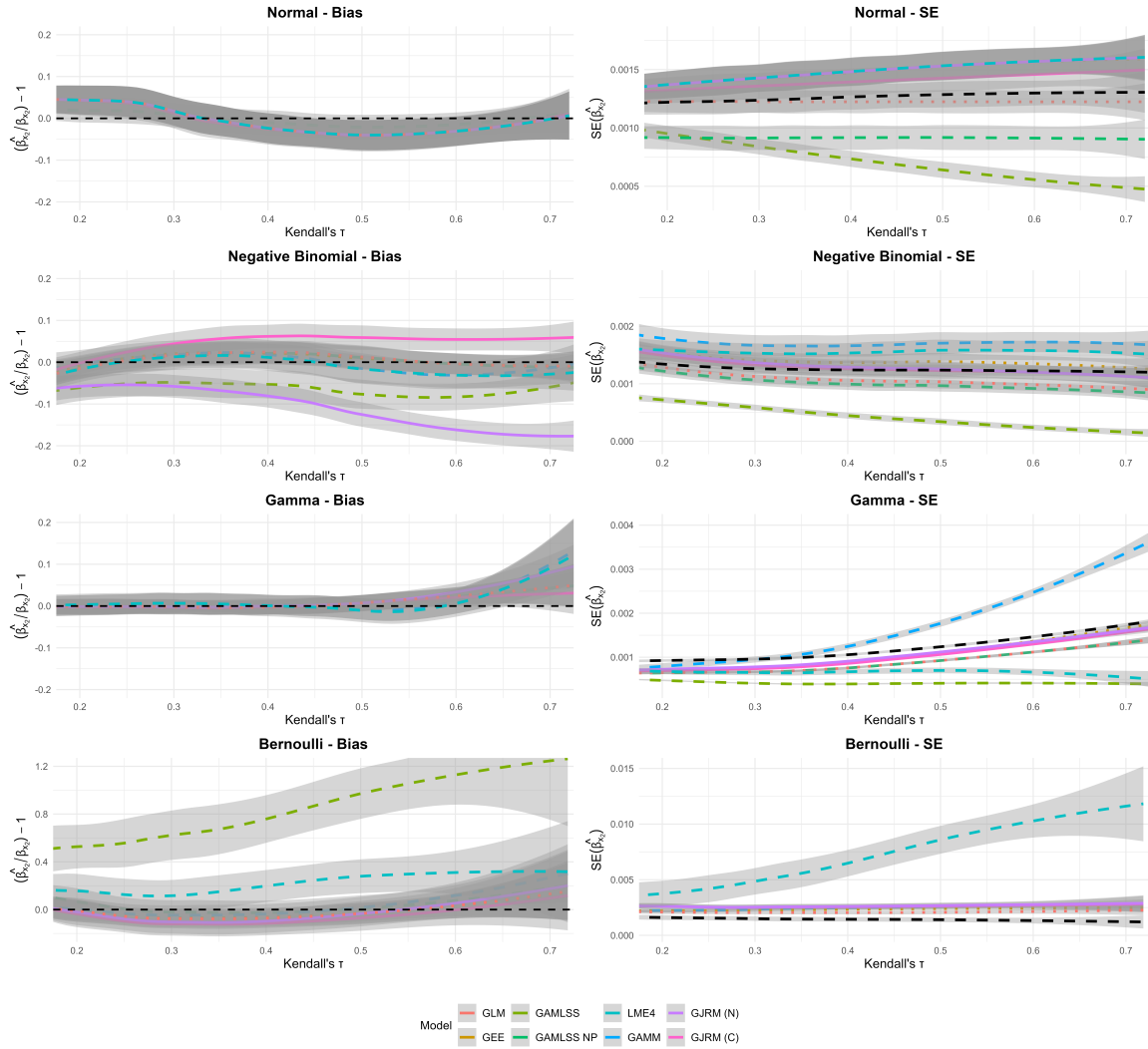


Figure 16: Bias for the estimate of the coefficient for x_2 (left) and its standard error (right) across the range of realizations of the simulated distributions, plotted against Kendall's τ . Random effect models (GLMMs) are shown as dashed lines, GJRMs are solid and GLM, GEE are dotted lines for ease of reference. Note the GAMLSS model results for the logistic are extremely variable with many extreme values and so cannot be shown entirely on the plot

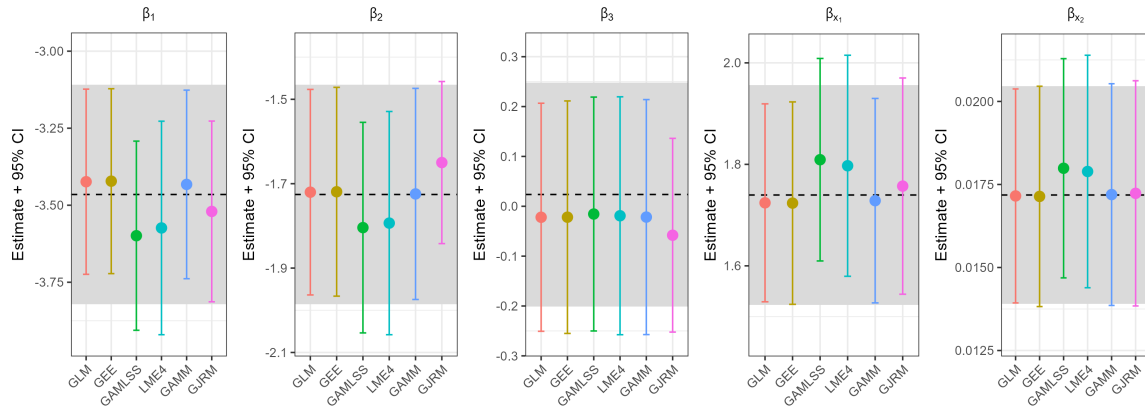


Figure 17: Trivariate Bernoulli; $\rho = 0.1$: fitted coefficients and 95 percent confidence intervals based on model estimates averaged over 100 simulations compared to the asymptotic coefficient (blue dotted line) and confidence interval.

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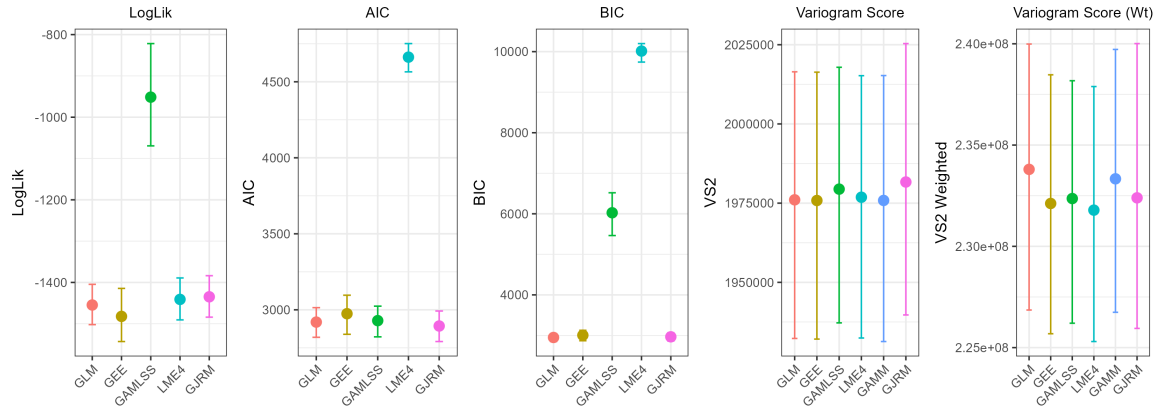


Figure 18: Trivariate Bernoulli; $\rho = 0.25$: model selection criteria based on model estimates averaged over 1000 simulations.

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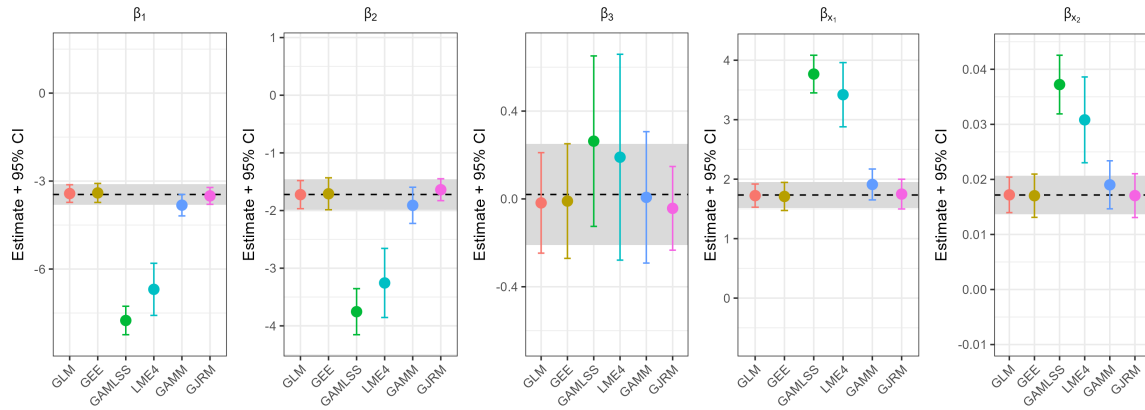


Figure 19: Trivariate Bernoulli; $\rho = 0.667$: fitted coefficients and 95 percent confidence intervals based on model estimates averaged over 1000 simulations compared to the asymptotic coefficient (black dotted line) and confidence interval.

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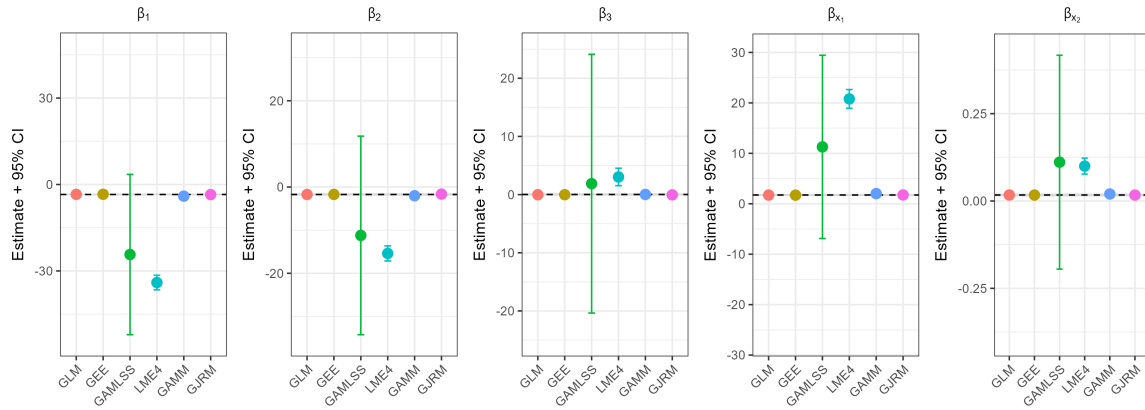


Figure 20: Trivariate Bernoulli; $\rho = 0.9$: fitted coefficients and 95 percent confidence intervals based on model estimates averaged over 100 simulations compared to the asymptotic coefficient (blue dotted line) and confidence interval.

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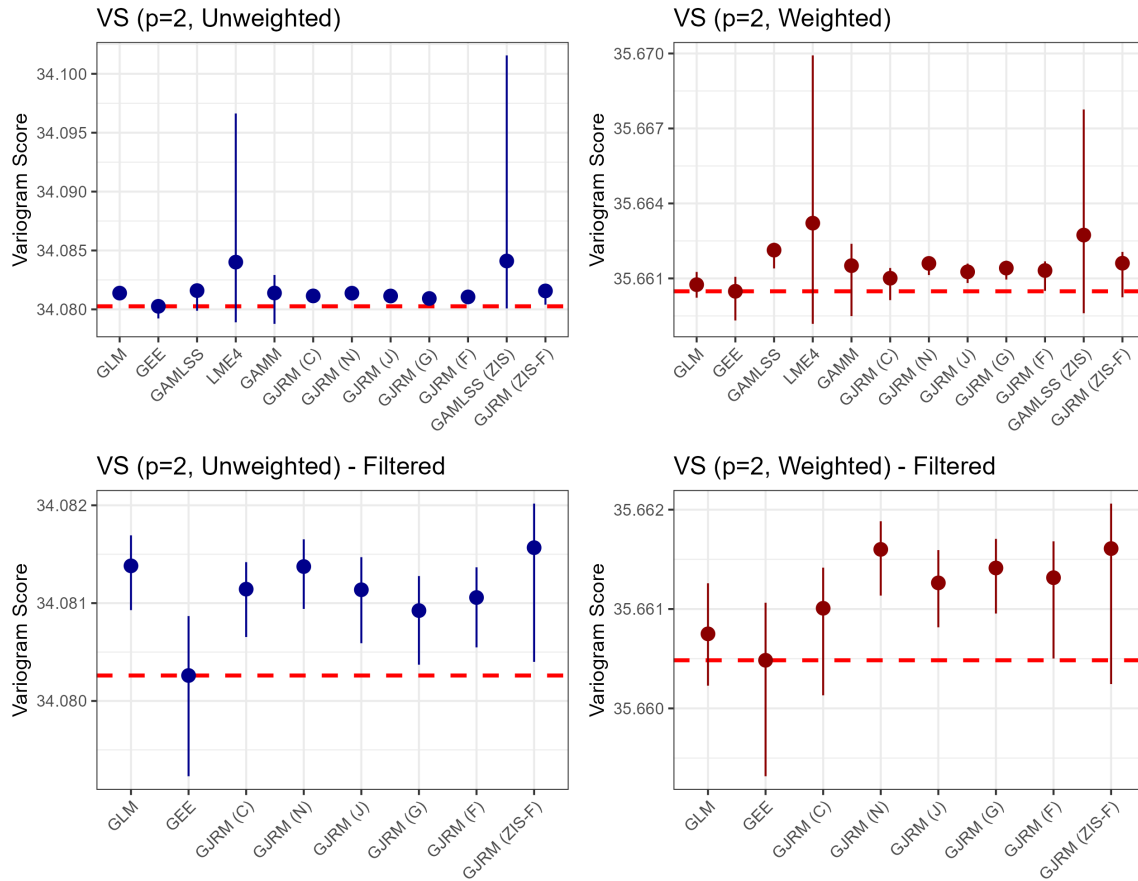


Figure 21: Weighted and unweighted variogram scores with $p = 2$ for the model fits to Doctor Visits data with variogram scores calculated over 100 simulations of 100 simulated realizations from each model. The lowest median value has been highlighted with a red dotted line for ease of comparison between models.

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